cft-anyons Lab Book

From category data to provably rigorous CFTs

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May 30, 2026

Contents

1 Frontmatter, North Star, and Claim Status

This is a reproducible research lab book. Its guiding goal — the *north star* — is a single, concrete construction problem:

> Given a (unitary) fusion or modular tensor category $\mathcal{C}$, together with whatever additional data is needed (OPE coefficients / conformal data), construct a family of microscopic (lattice) models and their symmetry generators whose continuum limit *provably* yields a mathematically rigorous QFT/CFT that (i) realises the input category data and (ii) carries a full projective unitary representation of the symmetries.

The working belief is that this is achievable at least for *rational* CFTs, and perhaps for all QFTs. This is a programme, not a settled theory: every step toward the goal is a question, proposal, or hypothesis until a local source, a local derivation, or a reproducible run supports it.

1.1 The Two Success Criteria

A construction is only a candidate solution if it meets both, non-negotiably:

1. **Provability.** The continuum limit is established rigorously (an operator-algebraic / wavelet renormalisation limit with theorems, not a heuristic scaling argument).
2. **Fidelity.** The realised continuum content faithfully reproduces the input category data $\mathcal{C}$, and the symmetry representation is *full* and projective-unitary.

1.2 Claim Status

No claim is treated as established until it is marked *Source-local* or *Checked* and linked to a local source, derivation, test, or run artifact. The status labels are fixed in `CONVENTIONS.md`: *Question*, *Proposal*, *Source-local*, *Checked*, and *Rejected*. These labels are part of the scientific record; they separate programme-level conjecture from results.

1.3 Evidence Chain

Every substantive assertion should be traceable through:

1. a local source manifest under `references/` (the kept papers are the ground truth),
2. a convention entry in `CONVENTIONS.md`,
3. a derivation, script, test, or run bundle when computation is involved,
4. an index row in `INDEX.md`, and
5. a report shard in this lab book.

Missing links are allowed during exploration only when they are explicit.

2 Programme Map

This shard is a map of questions, not a list of results. The north star (§1 / CA-00) factors into a pipeline of stages; each stage is an open problem with its own conventions, sources, and checkable invariants.

2.1 The Pipeline

$$\underbrace{\mathcal{C}\ (+\,\text{OPE/conformal data})}_{\text{input}} \longrightarrow \underbrace{\text{microscopic models}}_{\text{lattice Hilbert space + Hamiltonian}} \longrightarrow \underbrace{\text{lattice symmetry generators}}_{\text{e.g. Koo–Saleur } L_n}$$
$$\longrightarrow \underbrace{\text{provable continuum limit}}_{\text{OAR / wavelets}} \longrightarrow \underbrace{\text{rigorous (C)FT}}_{\text{conformal net / VOA / Wightman}}$$

2.2 Stage 1 — Category Input

- Which categories are in initial scope: multiplicity-free unitary fusion categories (Fibonacci, Ising/TL), then modular tensor categories?
- What auxiliary data beyond $\mathcal{C}$ is genuinely required (OPE coefficients, central charge, conformal weights), and for which target CFTs?
- Which gauge and indexing conventions are fixed before any F/R-symbol is written (see `CONVENTIONS.md`)?

2.3 Stage 2 — Microscopic Models

- Which family realises $\mathcal{C}$ on the lattice: anyon chains, string-nets, or another construction — and how is the family indexed by scale?
- What is the lattice Hilbert space (fusion-tree basis) and the Hamiltonian, and which invariants pin them (fusion-rule consistency, pentagon, projector idempotence)?

2.4 Stage 3 — Lattice Symmetry Generators

- Which lattice operators approximate the continuum symmetry generators (Koo–Saleur lattice Virasoro L_n, $\bar{L}_n$, and kin)?
- What convergence statement is the target, and what is checkable at finite size (commutator algebra, central charge extraction)?

2.5 Stage 4 — Provable Continuum Limit

- Which rigorous renormalisation framework carries the limit (operator- algebraic renormalisation, wavelet / multi-scale entanglement)?
- What must be proved: existence of the limit, the limiting Hilbert space, and convergence of the symmetry generators to a projective unitary representation?

2.6 Stage 5 — Rigorous (C)FT Target

- What is the target object: a conformal net, a vertex operator algebra, a Wightman theory, or an Osterwalder–Schrader theory?
- What is the fidelity statement: that the target's superselection / representation category recovers $\mathcal{C}$?

2.7 First Decisions

The first technical session should not start with formulas. It should register the sources for one worked input category (Fibonacci or Ising/TL), fix the minimum conventions (F/R-symbol gauge, vacuum index, fusion-tree ordering), and identify the smallest invariant that can be checked end-to-end on one stage of the pipeline.

3 Many-Particle Hilbert Spaces: the AND/OR/Vacuum Grammar

3.1 Sources and Dependencies

This is a pre-topic foundation for the programme (see CA-01, the programme map): how to build the Hilbert space of an indefinite number of particles of arbitrary types. The guiding idea is that two logical connectives map cleanly onto two Hilbert-space constructions, and the no-particle system is the complex line.

This shard treats the *finite* grammar and its semantics; the rigorous *indefinite*-particle (Hilbert-completed) case is CA-03. It harmonises with, and re-proves nothing from, the finite "AND/OR many-particle grammar" developed by the author in the sibling project (notation $\mathcal{V}$, $\mathcal{V}_n$, AND $= \otimes$, OR $= \oplus$, $\mathbf{1} = \mathbb{C}$); see that project's shard `AQM-70`. Statistics (Bose / Fermi / para) are deferred throughout.

3.2 The Three Primitives

Let particle *types* be (separable, complex) Hilbert spaces; a type's space is the state space of one particle of that type. Three primitives generate composite systems:

- **OR = direct sum $\oplus$.** "Either a particle of type 1 or of type 2" is $\mathcal{H}_1 \oplus \mathcal{H}_2$.
- **AND = tensor product $\otimes$.** "A particle of type 1 *and* a particle of type 2" is $\mathcal{H}_1 \otimes \mathcal{H}_2$.
- **vacuum = $\mathbb{C}$.** The no-particle system is the complex line.

Crucially, this grammar fixes *indefinite particle number*; exchange symmetry (statistics) is a *separate, later* step, realised as a quotient or subspace (CA-03, §"Symmetries come later"). Keeping the two apart is the whole point.

3.3 Two Monoidal Structures and Their Units

On the category Hilb of complex Hilbert spaces (with the obvious unitary coherence isomorphisms) there are two symmetric monoidal structures:

$$(\mathrm{Hilb},\ \oplus,\ \{0\}) \qquad \text{and} \qquad (\mathrm{Hilb},\ \otimes,\ \mathbb{C}),$$

with the two units *distinct*:

$$A \oplus \{0\} \ \cong\ A \qquad (\oplus\text{-unit is the } \textit{zero space } \{0\}),$$

$$A \otimes \mathbb{C} \ \cong\ A \qquad (\otimes\text{-unit is the } \textit{vacuum } \mathbb{C}).$$

The two are linked by a *distributivity* natural isomorphism, which on Hilbert spaces is *unitary*:

$$\delta:\ A \otimes (B \oplus C) \xrightarrow{\ \cong\ } (A \otimes B) \oplus (A \otimes C), \qquad A \otimes \{0\} \ \cong\ \{0\}.$$

Together $(\mathrm{Hilb}, \oplus, \{0\}, \otimes, \mathbb{C})$ is a symmetric *rig* (bimonoidal) category. Its coherence — that all diagrams built from associators, unitors, symmetries, and δ commute — is standard (Laplaza; Kelly) and is *not* re-proved here; downstream we use only the elementary, manifestly unitary associativity/unitor isomorphisms and the degree-0 identification $\mathbb{C} \otimes A \cong A$.

Vacuum = $\mathbb{C}$, not $\{0\}$. The vacuum is the $\otimes$-unit $\mathbb{C}$ (the empty tensor product, one particle's worth of "nothing"), a *one-dimensional* space carrying the unit vector Ω. It is *not* the $\oplus$-unit $\{0\}$ (the empty direct sum, the impossible system). Empty $\otimes$ is $\mathbb{C}$; empty $\oplus$ is $\{0\}$. Conflating them is the first pitfall this grammar guards against (cf. CONVENTIONS.md (a), which fixes the vacuum *object* as $\mathbb{C}$).

3.4 The Grammar and its Carrier

Fix a set $\{K_x\}_{x\in X}$ of single-particle type spaces (the *atoms*). Grammar expressions are

$$E ::= 0 \;\mid\; \mathbf{1} \;\mid\; K \;\mid\; \mathrm{AND}(E,F) \;\mid\; \mathrm{OR}(E,F),$$

where 0 is the zero atom, $\mathbf{1}$ is the vacuum atom ($\mathbf{1}=\mathbb{C}$), and K ranges over the atoms. The *carrier* $\mathcal{V}(E)$ is defined recursively:

$$\begin{gathered}\mathcal{V}(0)=\{0\}, \qquad\qquad \mathcal{V}(\mathbf{1})=\mathbb{C}, \qquad\qquad \mathcal{V}(K)=K,\\ \mathcal{V}(\mathrm{AND}(E,F))=\mathcal{V}(E)\otimes\mathcal{V}(F), \quad \mathcal{V}(\mathrm{OR}(E,F))=\mathcal{V}(E)\oplus\mathcal{V}(F).\end{gathered}$$

Thus AND is independent coexistence and OR is alternatives. Parenthesisation and order are part of the written expression; the rig coherence above guarantees that re-bracketing or re-ordering a *fixed* expression changes $\mathcal{V}(E)$ only by a canonical unitary isomorphism, so $\mathcal{V}$ is well defined up to canonical unitary isomorphism. (Morally $\mathcal{V}$ is the evaluation of the free symmetric rig category on $\{K_x\}$ into Hilb; we use this only as motivation, not as a load-bearing theorem.)

3.5 Particle-Number Grading

Declare each atom K to be *one* particle and $\mathbf{1}$ to be *zero* particles. The carrier then carries a particle-number grading $\mathcal{V}_n$:

$$\begin{gathered}\mathcal{V}_0(\mathbf{1})=\mathbb{C}, \quad \mathcal{V}_n(\mathbf{1})=\{0\}\ (n\neq 0), \qquad \mathcal{V}_1(K)=K, \quad \mathcal{V}_n(K)=\{0\}\ (n\neq 1), \quad \mathcal{V}_n(0)=\{0\},\\ \mathcal{V}_n(\mathrm{OR}(E,F))=\mathcal{V}_n(E)\oplus\mathcal{V}_n(F), \qquad \mathcal{V}_n(\mathrm{AND}(E,F))=\bigoplus_{i+j=n}\mathcal{V}_i(E)\otimes\mathcal{V}_j(F).\end{gathered}$$

For a *finite* expression the parse tree is finite, so only finitely many sectors are nonzero and $\mathcal{V}(E)=\bigoplus_{n\geq 0}\mathcal{V}_n(E)$ is a *finite* direct sum. The integer n is a grammar count of one-particle atoms; it is not a dynamics, a statistics rule, or (yet) an indefinite-particle space.

Worked atoms. A one-particle "menu" over a finite type set X is $P=\mathrm{OR}_{x\in X}K_x$, with $\mathcal{V}(P)=\bigoplus_{x\in X}K_x$ (the infinite, ℓ^2-completed menu is CA-03). The expression $\mathbf{1}\,\mathrm{OR}\,K$ has carrier $\mathbb{C}\oplus K$ — a "maybe" particle (either zero or one particle of type K), as in the sibling shard's $\mathbb{C}\oplus\mathbb{C}^d$. Tensoring N "maybe" atoms, $\mathrm{AND}_{j=1}^{N}(\mathbf{1}\,\mathrm{OR}\,K)$, has carrier $(\mathbb{C}\oplus K)^{\otimes N}$, whose grade-$n$ sector (by distributivity) is $\bigoplus_{|S|=n}\bigotimes_{x\in S}K$ over n-subsets of the N slots — the finite "occupation" picture.

3.6 Lamport Dependency Skeleton

$$\begin{aligned}
D1 &: \text{atoms } \{K_x\}_{x\in X} \text{ (single-particle Hilbert spaces); } \mathbf{1}=\mathbb{C} \text{ (vacuum)},\ 0=\{0\}.\\
D2 &: E ::= 0 \mid \mathbf{1} \mid K \mid \mathrm{AND}(E,F) \mid \mathrm{OR}(E,F).\\
D3 &: \mathcal{V}:\ \mathcal{V}(0)=\{0\},\ \mathcal{V}(\mathbf{1})=\mathbb{C},\ \mathcal{V}(\mathrm{AND})=\otimes,\ \mathcal{V}(\mathrm{OR})=\oplus.\\
D4 &: \mathcal{V}_0(\mathbf{1})=\mathbb{C},\ \mathcal{V}_1(K)=K;\ \mathcal{V}_n(\mathrm{OR})=\oplus,\ \mathcal{V}_n(\mathrm{AND})=\textstyle\bigoplus_{i+j=n}\mathcal{V}_i\otimes\mathcal{V}_j.\\
D5 &: (\mathrm{Hilb},\oplus,\{0\},\otimes,\mathbb{C}) \text{ a symmetric rig category, with}\\
&\quad\ \delta: A\otimes(B\oplus C)\xrightarrow{\cong}(A\otimes B)\oplus(A\otimes C) \text{ unitary and natural.}\\
T1 &: \text{(D5) coherence (Laplaza; Kelly)} \Rightarrow \mathcal{V} \text{ well defined up to canonical unitary iso.}\\
T2 &: \text{(D4) finite } E:\ \mathcal{V}(E)=\textstyle\bigoplus_{n\geq 0}\mathcal{V}_n(E) \text{ a finite sum (finite particle number).}
\end{aligned}$$

3.7 Scope and Deferral

Two extensions are deliberately out of this shard:

- **Indefinite particle number.** A genuine superposition over *unboundedly* many particle numbers has infinite support and does not live in the finite direct sum above; it requires the Hilbert (ℓ^2) completion. That is the content of CA-03, and is the new rigorous part of this pre-topic.
- **Statistics.** AND and OR choose no exchange symmetry. Bosonic / fermionic / para sectors arise later, by acting with the permutation group on tensor slots and passing to an invariant subspace or quotient (CA-03). This grammar only sets up the carrier those symmetries act on.

Pitfalls. (i) The $\oplus$-unit is $\{0\}$; the $\otimes$-unit (the vacuum) is $\mathbb{C}$ — distinct objects. (ii) Distributivity is an *isomorphism*, not an equality; all sector identities below hold up to canonical unitary iso. (iii) Everything here is *finite*; do not silently read the grading as an indefinite-particle space.

4 Indefinite Particle Number: the Full Fock Completion

4.1 Sources and Dependencies

This shard completes CA-02's finite AND/OR/vacuum grammar to a genuine *indefinite*-particle Hilbert space. It depends on CA-02 (the two monoidal structures, the units $\{0\}$ and $\mathbb{C}$, the carrier $\mathcal{V}$ and grading $\mathcal{V}_n$). The named Fock construction is anchored locally:

The sibling project's finite grammar (`AQM-70`) is recovered as the finite-truncation of the object built here.

4.2 Two Direct Sums — the Load-Bearing Distinction

Let $\{W_n\}_{n\geq 0}$ be a countable family of Hilbert spaces.

- The **algebraic direct sum** $\bigoplus_n^{\mathrm{alg}} W_n$ consists of finite-support sequences (w_n) ($w_n \in W_n$, $w_n = 0$ for all but finitely many n). It is a pre-Hilbert space, generally *incomplete*.
- The **Hilbert (ℓ^2) direct sum** $\widehat{\bigoplus}_n W_n$ consists of sequences with $\sum_n \|w_n\|^2_{W_n} < \infty$, with inner product $\langle (w_n), (v_n)\rangle = \sum_n \langle w_n, v_n\rangle_{W_n}$ (absolutely convergent by Cauchy–Schwarz). It is *complete*, hence a Hilbert space.

$\bigoplus_n^{\mathrm{alg}} W_n$ is a *dense* subspace of $\widehat{\bigoplus}_n W_n$, and the latter is its completion (standard functional analysis).

Why indefinite particle number forces $\widehat{\bigoplus}$. A state with nonzero amplitude in *unboundedly* many particle-number sectors (a genuine superposition over indefinitely many particles) has infinite support: it lies in $\widehat{\bigoplus}$ but not in $\bigoplus^{\mathrm{alg}}$. "Indefinite particle number" $\Leftrightarrow$ "not confined to finitely many sectors" $\Rightarrow$ the ℓ^2 completion is *required*. This is exactly what CA-02's finite grammar lacks.

4.3 Completed Tensor Powers

For Hilbert spaces H, K, the *completed* (Hilbert) tensor product $H \mathbin{\widehat{\otimes}} K$ is the completion of the algebraic tensor product under the inner product determined by $\langle a\otimes b,\, c\otimes d\rangle = \langle a, c\rangle_H \langle b, d\rangle_K$ (well defined and positive definite). For finite-dimensional factors $\widehat{\otimes} = \otimes$; the completion bites only for infinite-dimensional one-particle spaces. Define the *completed tensor powers*

$$H^{\widehat{\otimes} 0} := \mathbb{C} \;\; \text{(the vacuum sector)}, \qquad H^{\widehat{\otimes} n} := \underbrace{H \mathbin{\widehat{\otimes}} \cdots \mathbin{\widehat{\otimes}} H}_{n},$$

the bracketing immaterial up to the canonical unitary associativity isomorphism.

4.4 The Full Fock Space

Definition. For a one-particle Hilbert space H, the *full Fock space* is

$$\boxed{\mathfrak{F}(H) \;:=\; \widehat{\bigoplus}_{n\geq 0} H^{\widehat{\otimes} n}}\,, \qquad H^{\widehat{\otimes} 0} = \mathbb{C},$$

i.e. $\psi = (\psi_n)_{n\geq 0}$ with $\psi_n \in H^{\widehat{\otimes} n}$ and $\|\psi\|^2 = \sum_n \|\psi_n\|^2 < \infty$. The unit vector $\Omega = (1, 0, 0, \dots)$ in the degree-0 sector $\mathbb{C}$ is the *vacuum*. This is the *distinguishable* Fock space (the author's term in the local source `1910.10619`); in operator algebra it is also called the *free* or *Boltzmann* Fock space (standard terminology, not anchored in a local source), namely the Hilbert completion of the tensor algebra $T(H) = \bigoplus_n H^{\otimes n}$. It is *not* the symmetric (bosonic) or antisymmetric (fermionic) Fock space; see “Exchange sectors come later” below.

Compiler status. In the compiler grammar this is not an additional primitive constructor. It is the Hilbert completion of the ordinary tensor-power expressions $\mathbf{1}, H, H^{\widehat{\otimes}2}, \ldots$: a countable OR over all finite AND powers. If a one-particle symmetry representation is present, CA-05 makes this same object a graded direct-sum representation.

Theorem (Fock is a Hilbert space). $\mathfrak{F}(H)$ is a Hilbert space; (i) the algebraic direct sum $\bigoplus_n^{\mathrm{alg}} H^{\widehat{\otimes}n}$ (the finite-particle vectors) is *dense*; (ii) distinct-n sectors are *mutually orthogonal*, $\langle\psi,\varphi\rangle = \sum_n \langle\psi_n,\varphi_n\rangle$, so $H^{\widehat{\otimes}m} \perp H^{\widehat{\otimes}n}$ for $m \neq n$; (iii) on the degree-0 sector the inner product is the standard one on $\mathbb{C}$ and Ω is a unit vector. *Proof.* An instance of the ℓ^2-direct-sum facts above with $W_n = H^{\widehat{\otimes}n}$: completeness, density of the algebraic sum, and summand orthogonality are exactly those statements. □

The $1/N!$ caveat (a convention trap). Schottenloher's inner product on $D = \bigoplus_N H_N$ weights sector N by $1/N!$; that factor is a *symmetric*-Fock normalisation correcting the $N!$ overcounting of symmetric tensors. The full Fock space above uses the *plain* ℓ^2 sum with *no* combinatorial weight. Do not copy the $1/N!$ into the distinguishable case.

4.5 The Grammar Generates the Indefinite-Particle Space

Let the one-particle menu over a countable type set X be $P = \mathrm{OR}_{x\in X} K_x$, with carrier $\mathcal{V}(P) = \widehat{\bigoplus}_{x\in X} K_x$ (the ℓ^2 direct sum; an ordinary finite direct sum when X is finite). The indefinite-particle space over this menu is $\mathfrak{F}(\mathcal{V}(P))$. Expanding each completed tensor power by distributivity (CA-02's δ, extended unitarily to the completions) gives the $(n, \text{type-word})$-graded decomposition

$$\mathcal{V}(P)^{\widehat{\otimes}n} \cong \widehat{\bigoplus}_{w\in X^n} K_w, \quad K_w := K_{x_1} \widehat{\otimes} \cdots \widehat{\otimes} K_{x_n}, \qquad \Longrightarrow \qquad \mathfrak{F}(\mathcal{V}(P)) \cong \widehat{\bigoplus}_{n\geq 0} \widehat{\bigoplus}_{w\in X^n} K_w.$$

Each of the n tensor slots independently chooses a type $x_i \in X$ (two slots may choose the same type; nothing is symmetrised). The grading is $\mathcal{V}_n = \widehat{\bigoplus}_{w\in X^n} K_w$ (n = particle number, $w \in X^n$ = ordered type-word) — the ℓ^2-completed, all-n analogue of CA-02's finite grade-n sector. The sibling project's finite expressions ($P_X^{\mathrm{AND}\, m}$, the truncation $T_{\leq M}$) are precisely the finite-support / finite-M restrictions of this object; that is the exact harmonisation: `AQM-70` is the $\bigoplus^{\mathrm{alg}}$/finite-M truncation of $\mathfrak{F}$.

4.6 Exchange Sectors Come Later

The full Fock space already carries symmetry representations when H does: internal symmetries act sectorwise by tensor powers (CA-05), and for each n the permutation group S_n acts unitarily on $H^{\widehat{\otimes}n}$ by permuting tensor slots. What is deferred is the *sector selection*. Bosonic and fermionic Fock spaces are the images of the per-sector orthogonal projectors

$$\Gamma_{\mathrm{s}}(H) = P_{\mathrm{sym}}\,\mathfrak{F}(H) \quad \text{(symmetric / Bose)}, \qquad \Gamma_{\mathrm{a}}(H) = P_{\mathrm{anti}}\,\mathfrak{F}(H) \quad \text{(antisymmetric / Fermi)},$$

with para-statistics the other Young-symmetry isotypic components. This is the “impose an equivalence relation under particle exchange” step. We define no projector and prove nothing here:

the construction of these symmetry quotients/subspaces, and the genuinely anyonic case (where standard Fock space is not the right object), are deferred to later shards.

4.7 Lamport Dependency Skeleton

$$\begin{aligned}
&D6: \bigoplus^{\mathrm{alg}}(\text{finite support}) \;\subset\; \widehat{\bigoplus}(\ell^2): \;\langle (w_n),(v_n)\rangle = \textstyle\sum_n \langle w_n, v_n\rangle.\\
&D7: \text{completed tensor power } H^{\widehat{\otimes} n},\; H^{\widehat{\otimes} 0} = \mathbb{C} \text{ (vacuum)}.\\
&D8: \mathfrak{F}(H) := \widehat{\bigoplus}_{n\geq 0} H^{\widehat{\otimes} n} \text{ (full / distinguishable Fock)}; \Omega = (1,0,\dots).\\
&D9: P = \mathrm{OR}_{x\in X} K_x; \quad \mathfrak{F}(\mathcal{V}(P)) = \widehat{\bigoplus}_{n\geq 0} \mathcal{V}(P)^{\widehat{\otimes} n}.\\
&T3: \text{(D6,D7) } \mathfrak{F}(H) \text{ Hilbert}; \bigoplus^{\mathrm{alg}} \text{ dense}; H^{\widehat{\otimes} m} \perp H^{\widehat{\otimes} n}\ (m \neq n).\\
&T4: \text{indefinite particle no.} \Leftrightarrow \text{infinite support} \Rightarrow \widehat{\bigoplus} \text{ (not } \bigoplus^{\mathrm{alg}}).\\
&T5: (\text{D8},\delta)\ \mathfrak{F}(\mathcal{V}(P)) \cong \widehat{\bigoplus}_{n\geq 0} \widehat{\bigoplus}_{w\in X^n} K_w \text{ ((}n, \text{type-word) grading)}.\\
&T6: \Gamma_{\mathrm{s}} = P_{\mathrm{sym}}\mathfrak{F},\; \Gamma_{\mathrm{a}} = P_{\mathrm{anti}}\mathfrak{F} \text{ deferred (statistics later)}.
\end{aligned}$$

4.8 Scope and Pitfalls

In scope: the indefinite-particle direction, i.e. countable OR (the Fock completion). **Out of scope:** countable AND (an *infinite tensor product*), which is analytically a different object (von Neumann's infinite tensor product, needing a stabilising reference sequence) — not developed here.

Pitfalls: (i) *algebraic* $\bigoplus^{\mathrm{alg}}$ (finite particle number, dense, incomplete) vs. *completed* $\widehat{\bigoplus}$ (indefinite, complete) — the distinction this shard exists to make. (ii) For infinite-dimensional one-particle H, the AND of CA-02 must mean the *completed* $\widehat{\otimes}$, not the algebraic $\otimes$. (iii) The $1/N!$ symmetric-Fock weight must not be imported into the distinguishable case. (iv) All sector identities hold on dense subspaces and extend by continuity; states are ℓ^2-summable and cross-sector inner products vanish.

5 A Hilbert-Space Compiler Contract

5.1 Status and Local Anchors

Status: Proposal / contract shard. This shard introduces no new mathematical classification theorem. It records a design target for the pre-work sequence around CA-02 and CA-03: a typed grammar of many-particle expressions should be compilable to Hilbert-space data and observable-algebra data with explicit evidence status.

Local anchors are CA-02 for the finite AND/OR/vacuum grammar, CA-03 for the Hilbert direct-sum Fock completion, and `CONVENTIONS.md` entries (f) and (g) for symmetry and sector semantics. The first executable check for this sequence is the finite averaging projector in `test/runtests.jl`, backed by `src/CftAnyons.jl`.

5.2 Mini North Star

Proposal: build a *Hilbert-space compiler*. The user supplies an expression in a typed grammar, together with the atom table and required policies; the compiler returns the Hilbert carrier, its observable algebra policy, its symmetry representation data, and any sector projection or quotient presentation that was requested.

The motivating analogy is "regular expressions for quantum mechanics": a small syntax has compositional semantics. The analogy is only a prompt. Here the syntax is typed and semantics-bearing; every constructor must say what it does to Hilbert carriers, particle gradings, observable algebras, symmetries, and evidence metadata.

5.3 Input Record

A compiled expression is not just a string. It has an input record:

$$\mathsf{Input} = (\mathsf{Atoms},\ E,\ \mathsf{Completion},\ \mathsf{ObsPolicy},\ \mathsf{Symmetry},\ \mathsf{SectorPolicy}).$$

Atoms**.** A table of atoms. At minimum, each atom names a Hilbert space K_x. Later entries may include a group representation, a particle degree, a source locator, or category data.

E**.** A grammar expression built from $0, \mathbf{1}, K_x, \mathrm{AND}, \mathrm{OR}$, and later sector constructors. Fock-style all-n notation is treated as a derived completion of tensor powers, not as a primitive grammar constructor.

$\mathsf{Completion}$**.** Whether $\oplus, \otimes$ are finite Hilbert operations, Hilbert completions $\widehat{\bigoplus}, \widehat{\otimes}$, or a finite cutoff. CA-03 fixes the first infinite completion of the derived all-tensor-power construction used here.

$\mathsf{ObsPolicy}$**.** The observable-algebra choice. There is no silent default: full $\mathcal{B}(H)$, block-preserving, tensor-local, symmetry-invariant, and compressed sector algebras are different outputs.

$\mathsf{Symmetry}$**.** Optional unitary or projective-unitary representation data, handled according to CONVENTIONS.md (f).

$\mathsf{SectorPolicy}$**.** Optional projection, subspace, or quotient data; see CONVENTIONS.md (g).

5.4 Output Record

The intended output is a typed record, not only a Hilbert-space dimension:

$$\mathsf{Compile}(E) = (H_E,\ \{H_{E,n}\}_{n\geq 0},\ \mathcal{A}_E,\ U_E,\ P_E,\ Q_E,\ \mathsf{Evidence}_E).$$

H_E. The Hilbert carrier. CA-02 and CA-03 define the first carrier semantics.

$\{H_{E,n}\}$. The particle-number grading when the atom table declares particle degrees.

$\mathcal{A}_E$. The selected observable algebra. This slot is mandatory once any observable claim is made.

U_E. The propagated symmetry representation, if symmetry data were supplied.

P_E. The selected sector projection, if a sector constructor was used.

Q_E. The quotient presentation, if one was requested and its closed relation subspace was named.

$\mathsf{Evidence}_E$. Source-local, checked, proposal, question, or rejected evidence tags for the pieces above.

5.5 Observable-Algebra Policies

The carrier alone does not determine the observable algebra relevant to a model. For example, $H \oplus K$ supports at least:

$$\mathcal{B}(H \oplus K), \qquad \mathcal{B}(H) \oplus \mathcal{B}(K), \qquad \text{off-diagonal creation/annihilation blocks between } H \text{ and } K.$$

All three are legitimate in different contexts; none is forced by the word "OR" alone. Similarly, $H \otimes K$ supports the full algebra $\mathcal{B}(H \otimes K)$ and the tensor-local subalgebra generated by $\mathcal{B}(H) \otimes I$ and $I \otimes \mathcal{B}(K)$.

Compiler rule: no report statement may infer an observable algebra merely from a carrier constructor. The statement must name the observable policy, or remain a carrier-only statement.

5.6 Lamport Dependency Skeleton

$$\begin{aligned}
D10 &: \mathsf{Input} = (\mathsf{Atoms}, E, \mathsf{Completion}, \mathsf{ObsPolicy}, \mathsf{Symmetry}, \mathsf{SectorPolicy}).\\
D11 &: \mathsf{Compile}(E) = (H_E, \{H_{E,n}\}, \mathcal{A}_E, U_E, P_E, Q_E, \mathsf{Evidence}_E).\\
D12 &: \mathsf{ObsPolicy} \in \{\text{carrier-only}, \mathcal{B}(H), \text{block}, \text{tensor-local}, \text{invariant}, \text{compressed}\}.\\
T7 &: \text{A carrier claim needs } H_E;\ \text{an observable claim also needs } \mathcal{A}_E.\\
T8 &: \text{The compiler analogy is a proposal, not a theorem of canonical quantization.}
\end{aligned}$$

5.7 Boundary

This contract does not yet compile fusion-category labels, anyonic braid statistics, Hamiltonians, continuum limits, or CFT data. It only fixes the pre-work target: future shards should be able to say exactly which input record they compile and which output slots they have actually justified.

6 Symmetry-Decorated Many-Particle Grammar

6.1 Sources and Dependencies

This shard extends CA-02 and CA-03 by decorating atoms with symmetry data. It uses CONVENTIONS.md (f).

6.2 Decorated Atoms

Fix a topological group G. A symmetry-decorated atom is a pair

$$(K_x, U_x), \qquad U_x : G \to U(K_x),$$

where K_x is the atomic Hilbert space and U_x is a unitary representation. If the intended symmetry is projective, the pre-work convention is to choose a central extension

$$1 \to U(1) \to E \to G \to 1$$

and use an honest unitary representation $U_x : E \to U(K_x)$. The expression is then compiled over E, with a note that the induced physical symmetry is projective over G.

The vacuum atom is $(\mathbb{C}, \mathbf{1})$, where $\mathbf{1}(g)z = z$. The zero atom carries the unique representation on $\{0\}$.

6.3 Propagation Through OR and AND

If E, F have compiled carriers and representations (H_E, U_E) and (H_F, U_F) for the same group or chosen central extension, define

$$H_{\mathrm{OR}(E,F)} = H_E \oplus H_F, \qquad U_{\mathrm{OR}(E,F)}(g) = U_E(g) \oplus U_F(g),$$

and

$$H_{\mathrm{AND}(E,F)} = H_E \mathbin{\widehat{\otimes}} H_F, \qquad U_{\mathrm{AND}(E,F)}(g) = U_E(g) \mathbin{\widehat{\otimes}} U_F(g).$$

For finite-dimensional factors, $\widehat{\otimes} = \otimes$. In infinite-dimensional Hilbert spaces, the completed tensor product is the one already required by CA-03.

These formulas are the Hilbert-space version of the monoidal structure on $\mathrm{Rep}^\dagger(G)$: direct sums are block actions and tensor products are diagonal actions. If two atoms use different groups, different central extensions, or incompatible projective multipliers, the compiler must stop and ask for a common symmetry object before forming a single U_E.

6.4 The Derived Full-Fock Representation

For an ordinary unitary representation $U : G \to U(H)$, the full-Fock object is itself a graded representation:

$$\mathfrak{F}(U)(g) = \bigoplus_{n \geq 0} U(g)^{\widehat{\otimes} n} \quad \text{on} \quad \mathfrak{F}(H) = \widehat{\bigoplus}_{n \geq 0} H^{\widehat{\otimes} n},$$

with $U(g)^{\widehat{\otimes} 0} = \mathrm{id}_{\mathbb{C}}$. This preserves the particle-number grading. This is not a new grammar constructor: it is the direct sum of the tensor-power representations already produced by OR and AND, completed as in CA-03. In this pre-work shard, no Shale–Stinespring-type implementability theorem is claimed for general canonical field algebras. Implementing a one-particle unitary on a chosen CAR/CCR observable algebra is a separate field-algebra question.

6.5 Intertwiners and Symmetry-Preserving Compilation

A morphism between decorated atoms or compiled expressions is symmetry-preserving only when it is an intertwiner:

$$TU_E(g) = U_F(g)T \quad \text{for all } g.$$

This matters for compiler optimisations. Rebracketing and distributivity isomorphisms used by CA-02 are acceptable in the decorated grammar only when they also intertwine the propagated actions. The standard Hilbert-space associators, unitors, symmetries, and distributivity maps do so for diagonal direct-sum and tensor-product actions.

6.6 Lamport Dependency Skeleton

$D13$: decorated atom (K_x, U_x), $U_x : G \to U(K_x)$.
$D14$: projective symmetry over G is compiled via $U_x : E \to U(K_x)$ for a named central extension E.
$D15$: $U_{E\,\mathrm{OR}\,F} = U_E \oplus U_F, \quad U_{E\,\mathrm{AND}\,F} = U_E \mathbin{\widehat{\otimes}} U_F$.
$D16$: $\mathfrak{F}(U) = \widehat{\bigoplus}_{n\geq 0} U^{\widehat{\otimes} n}, \quad U^{\widehat{\otimes} 0} = 1_{\mathbb{C}}$.
$T9$: The propagated symmetry preserves the CA-02/CA-03 particle grading.
$T10$: A single compiled U_E requires a common group or central extension.

6.7 Boundary

This shard does not yet impose invariance, select charge sectors, form quotient spaces, or define symmetry-invariant observable algebras. Those are sector operations, not mere symmetry propagation, and are treated in CA-06.

7 Sectors, Projections, Subspaces, and Quotients

7.1 Sources and Dependencies

This shard depends on CA-04 and CA-05 and fixes the first sector semantics for the Hilbert-space compiler. It uses CONVENTIONS.md (g).

7.2 The Triangle

Let H be a Hilbert space. The compiler treats a sector first as an orthogonal projection

$$P = P^* = P^2 \in \mathcal{B}(H).$$

It may then expose either of the equivalent Hilbert-space presentations

$$H_P := \operatorname{Ran}(P) \subset H, \qquad H/\ker(P) = H/\operatorname{Ran}(1-P).$$

The map

$$[v] \in H/\ker(P) \longmapsto Pv \in \operatorname{Ran}(P)$$

is a well-defined unitary isomorphism when the quotient has the Hilbert quotient norm. Thus the safe slogan is not "symmetry equals quotient" but:

$$\text{projection} \quad \Longleftrightarrow \quad \text{closed subspace} \quad \Longleftrightarrow \quad \text{quotient by the named orthogonal complement.}$$

7.3 Finite-Group Invariants

Let G be finite and $U : G \to U(H)$ a unitary representation. Define

$$P_G = \frac{1}{|G|} \sum_{g \in G} U(g).$$

Then P_G is the orthogonal projection onto the invariant subspace

$$H^G = \{v \in H : U(g)v = v \ \forall g \in G\}.$$

Proof. Idempotence follows by reindexing:

$$P_G^2 = \frac{1}{|G|^2} \sum_{g,h} U(gh) = \frac{1}{|G|} \sum_{k} U(k) = P_G.$$

Self-adjointness follows from unitarity and $g \mapsto g^{-1}$:

$$P_G^* = \frac{1}{|G|} \sum_{g} U(g)^* = \frac{1}{|G|} \sum_{g} U(g^{-1}) = P_G.$$

For every k, $U(k)P_G = P_G$, so $\operatorname{Ran}(P_G) \subset H^G$. If $v \in H^G$, then $P_G v = v$, so $H^G \subset \operatorname{Ran}(P_G)$. $\square$

7.4 Coinvariant Quotient

For the same finite action, let

$$R_G = \operatorname{span}\{U(g)v - v : g \in G,\ v \in H\}.$$

Then $R_G \subseteq \ker(P_G)$ because $P_G U(g) = P_G$. Conversely, if $w \in \ker(P_G)$, then

$$w = \frac{1}{|G|} \sum_{g \in G} \big(w - U(g)w\big) \in R_G.$$

Therefore $R_G = \ker(P_G) = (H^G)^{\perp}$, and the coinvariant quotient

$$H_G := H/R_G$$

is canonically unitarily isomorphic to the invariant subspace H^G. For compact non-finite groups the analogous statement requires a Haar-integral and analytic hypotheses; this shard records only the finite checked case.

7.5 Observable-Algebra Distinctions

There are three different symmetry-related algebras that the compiler must not identify silently:

$$\mathcal{B}(H), \qquad \mathcal{B}(H)^G := \{A : U(g)AU(g)^* = A\ \forall g\}, \qquad P_G \mathcal{B}(H) P_G \cong \mathcal{B}(H^G).$$

The first is the full algebra before imposing symmetry. The second is the algebra of invariant observables on the original carrier. The third is the compressed algebra acting on the invariant sector. A model may choose one of these, but the choice is an observable policy in the sense of CA-04, not a consequence of the word "symmetry" alone.

7.6 Checked Example

The current Julia check uses $G = C_2$ acting on $\mathbb{C}^2$ by the swap matrix S. The averaging projection is

$$P = \frac{1}{2}(I + S) = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{pmatrix}.$$

The test checks $P^* = P = P^2$, checks $I - P$ is the complementary orthogonal projection, verifies $P(I - P) = 0$, and verifies that $(1, 1)$ is fixed while $(1, -1)$ is killed. This is the first executable invariant for the sector compiler.

7.7 Lamport Dependency Skeleton

$$
\begin{aligned}
D17 &: \text{sector projection } P = P^* = P^2; \quad H_P = \operatorname{Ran}(P). \\
D18 &: H/\ker(P) \xrightarrow{[v]\mapsto Pv} \operatorname{Ran}(P). \\
D19 &: P_G = |G|^{-1} \textstyle\sum_g U(g) \quad (G \text{ finite}). \\
D20 &: R_G = \operatorname{span}\{U(g)v - v\}. \\
L1 &: P_G = P_G^* = P_G^2, \quad \operatorname{Ran}(P_G) = H^G. \\
L2 &: R_G = \ker(P_G) = (H^G)^{\perp}, \quad H/R_G \cong H^G. \\
T11 &: \text{Projection, subspace, and quotient are equivalent only after the relation subspace is named.} \\
T12 &: \mathcal{B}(H)^G \text{ and } P_G \mathcal{B}(H) P_G \text{ are different observable policies.}
\end{aligned}
$$

7.8 Boundary

The finite averaging theorem is not a statement about spontaneous symmetry breaking, gauge constraints, BRST cohomology, braid statistics, or superselection sectors in a continuum CFT. It is the first Hilbert-space sector rule needed by the compiler.

8 Exchange Symmetry and Statistics as Sector Selection

8.1 Sources and Dependencies

This shard discharges part of the deferral in CA-03: ordinary Bose/Fermi statistics are treated as choices of irreducible exchange-symmetry sectors on the full distinguishable Fock carrier. It depends on CA-05 for internal symmetry propagation and CA-06 for projection/subspace/quotient semantics.

8.2 Two Symmetry Layers

There are two distinct actions on many-particle carriers:

$$\text{internal symmetry } G, \qquad \text{exchange symmetry } S_n.$$

If $U : G \to U(H)$ is an internal one-particle representation, then on the n-particle distinguishable sector $H^{\widehat{\otimes} n}$ it acts diagonally:

$$U_n(g) = U(g)^{\widehat{\otimes} n}.$$

The permutation group S_n acts by permuting tensor slots:

$$\Pi_n(\sigma)(v_1 \otimes \cdots \otimes v_n) = v_{\sigma^{-1}(1)} \otimes \cdots \otimes v_{\sigma^{-1}(n)}.$$

The two actions commute:

$$\Pi_n(\sigma) U_n(g) = U_n(g) \Pi_n(\sigma).$$

Thus internal charge sectors and exchange-statistics sectors can be imposed in either order in the ordinary tensor-slot setting.

8.3 Irrep-Labeled Exchange Projectors

The representation $\Pi_n : S_n \to U(H^{\widehat{\otimes} n})$ is the primary datum. Sector projections are supplied by irreducible representation data of S_n, not by specialised grammar constructors. For a one-dimensional unitary character $\chi : S_n \to U(1)$, CA-06 applied to the twisted representation $\overline{\chi}\,\Pi_n$ gives

$$P_{\chi,n} = \frac{1}{n!} \sum_{\sigma \in S_n} \overline{\chi(\sigma)}\,\Pi_n(\sigma),$$

whose range is the χ-sector

$$\{w \in H^{\widehat{\otimes} n} : \Pi_n(\sigma) w = \chi(\sigma) w \ \forall \sigma \in S_n\}.$$

For the trivial character 1, this is the symmetric projector:

$$P_{\mathrm{s},n} = \frac{1}{n!} \sum_{\sigma \in S_n} \Pi_n(\sigma),$$

and for the sign character sgn, this is the antisymmetric projector:

$$P_{\mathrm{a},n} = \frac{1}{n!} \sum_{\sigma \in S_n} \mathrm{sgn}(\sigma) \Pi_n(\sigma).$$

Their ranges are the symmetric and antisymmetric tensor powers:

$$\mathrm{Ran}(P_{\mathrm{s},n}) = H^{\widehat{\otimes}_s n}, \qquad \mathrm{Ran}(P_{\mathrm{a},n}) = \bigwedge^n H.$$

The corresponding Fock sectors are

$$\Gamma_{\mathrm{s}}(H) = \widehat{\bigoplus}_{n\geq 0} \mathrm{Ran}(P_{\mathrm{s},n}), \qquad \Gamma_{\mathrm{a}}(H) = \widehat{\bigoplus}_{n\geq 0} \mathrm{Ran}(P_{\mathrm{a},n}).$$

Thus "Bose" and "Fermi" name the trivial- and sign-irrep choices for ordinary exchange symmetry. The compiler should record the selected irrep/character and the resulting projection, not a special Bose/Fermi syntax node.

8.4 Quotient Presentations

For a one-dimensional character χ, the sector can also be presented as a quotient of the full tensor algebra by the closed linear span of relations

$$\Pi_n(\sigma)v - \chi(\sigma)v.$$

The trivial character recovers symmetric relations; the sign character recovers antisymmetric signed relations. In the compiler, these quotient presentations are secondary: the primary sector object is the orthogonal projector, because it fixes the inherited Hilbert structure and the compressed observable algebra.

8.5 Para and Anyonic Boundaries

Higher-dimensional irreducible S_n-sectors require central idempotents or matrix-block projectors for the group algebra of S_n. This is exactly the same representation/irrep principle, but this shard records it as a proposal rather than a completed construction; a later shard must source the convention and projector normalization before using it.

Genuinely anyonic exchange is not an S_n-projection problem on ordinary tensor slots. It involves braid-group or categorical data and, in this project, must be handled together with the fusion-category conventions for associators, braiding, and F/R-symbols. No braid-statistics projector is defined here.

8.6 Lamport Dependency Skeleton

$$\begin{aligned}
D21 &: U_n(g) = U(g)^{\widehat{\otimes} n} \quad \text{(internal symmetry on } H^{\widehat{\otimes} n}\text{)}.\\
D22 &: \Pi_n : S_n \to U(H^{\widehat{\otimes} n}) \quad \text{permutes tensor slots.}\\
D23 &: P_{\chi,n} = n!^{-1} \textstyle\sum_\sigma \overline{\chi(\sigma)}\Pi_n(\sigma) \quad (\chi \text{ a one-dimensional } S_n \text{ character}).\\
L3 &: \Pi_n(\sigma)U_n(g) = U_n(g)\Pi_n(\sigma).\\
L4 &: P_{\chi,n} \text{ is an orthogonal projection by CA-06 finite averaging.}\\
T13 &: \Gamma_{\mathrm{s}}, \Gamma_{\mathrm{a}} \text{ are the trivial/sign irrep sectors inside full Fock.}\\
T14 &: \text{Anyonic exchange remains deferred to categorical braid data.}
\end{aligned}$$

8.7 Compiler Consequence

The compiler now has its first nontrivial sector family: given the exchange representation $\Pi_n : S_n \to U(H^{\widehat{\otimes} n})$, a selected irrep character χ supplies $P_{\chi,n}$, its range, its quotient relations, and the chosen compressed observable algebra. There is no primitive Bose, Fermi, or Fock grammar node in this account; Fock is the derived graded representation of CA-05, and Bose/Fermi are names for particular irrep choices of exchange symmetry.

9 Scoping the Hilbert-Space Compiler

9.1 Status, Sources, and Dependencies

Status: specification shard. This shard turns CA-04's contract into a first partial compiler specification. Its scope is the pre-work grammar of CA-02–CA-07: finite Hilbert carriers, derived all-tensor-power completions, unitary/projective symmetries via central extensions, and irrep/projection sectors.

9.2 Compile Judgement

The compiler is a *partial* map. A successful judgement has the form

$$\Gamma; \Pi \vdash E \Downarrow (H_E, \{H_{E,n}\}, \mathcal{A}_E, U_E, P_E, Q_E, \mathsf{Ev}_E),$$

where Γ is the atom table and Π is the policy table. A failed judgement has the form

$$\Gamma; \Pi \vdash E \Uparrow \mathsf{reason}.$$

The failure result is part of the specification: incompatible symmetry data, missing observable policy, or an ill-typed sector constructor must stop compilation.

The atom table entry for x has at least a Hilbert space K_x, a degree $d_x \in \mathbb{N}$ (usually 1), evidence metadata, and optionally symmetry data $U_x : G \to U(K_x)$ or a named projective central extension as in CA-05.

9.3 Surface Language for the First Compiler

The first surface language is deliberately small:

$$E ::= 0 \mid \mathbf{1} \mid x \mid \mathrm{OR}(E, F) \mid \mathrm{AND}(E, F) \mid \mathrm{Sector}_{G,\chi}(E).$$

Here G acts on the compiled carrier and χ is a one-dimensional unitary character; invariants are the case $\chi = 1$. Fock, finite truncation, symmetric sectors, and antisymmetric sectors are derived from OR/AND, representation propagation, and the generic sector operation.

For a one-particle atom x, define tensor-power expressions recursively by

$$x^{[0]} := \mathbf{1}, \qquad x^{[m+1]} := \mathrm{AND}(x^{[m]}, x).$$

Then

$$\mathsf{T}_{\le M}(x) := \mathrm{OR}_{m=0}^{M} x^{[m]}, \qquad \mathsf{T}_\infty(x) := \text{Hilbert completion of } \mathrm{OR}_{m\ge 0} x^{[m]}.$$

$\mathsf{T}_\infty(x)$ is the usual full/distinguishable Fock space $\widehat{\bigoplus}_{m\ge 0} K_x^{\widehat{\otimes} m}$, but as a derived graded representation.

9.4 Constructor Rules

The carrier, grading, and symmetry rules are recursive:

$$H_0 = \{0\}, \quad H_{\mathbf{1}} = \mathbb{C}, \quad H_x = K_x,$$

$$H_{\mathrm{OR}(E,F)} = H_E \oplus H_F, \qquad H_{\mathrm{AND}(E,F)} = H_E \mathbin{\widehat{\otimes}} H_F.$$

The grading rules are CA-02's:

$$H_{\mathrm{OR}(E,F),n} = H_{E,n} \oplus H_{F,n}, \qquad H_{\mathrm{AND}(E,F),n} = \bigoplus_{i+j=n} H_{E,i} \mathbin{\widehat{\otimes}} H_{F,j}.$$

If U_E, U_F are representations of the same group or named central extension, the symmetry rules are

$$U_{\mathrm{OR}}(g) = U_E(g) \oplus U_F(g), \qquad U_{\mathrm{AND}}(g) = U_E(g) \mathbin{\widehat{\otimes}} U_F(g).$$

If the symmetry objects are not the same, the carrier may still compile, but the symmetry output is rejected unless the requested output was explicitly carrier-only.

For the derived tensor-power closures,

$$H_{\mathsf{T}_\infty(x)} = \widehat{\bigoplus}_{m\geq 0} K_x^{\widehat{\otimes} m}, \qquad H_{\mathsf{T}_{\leq M}(x)} = \bigoplus_{m=0}^{M} K_x^{\otimes m}.$$

For finite examples the compiler may report dimensions:

$$\begin{aligned} \dim \mathrm{OR} &= \dim H_E + \dim H_F, \\ \dim \mathrm{AND} &= (\dim H_E)(\dim H_F), \\ \dim \mathsf{T}_{\leq M}(x) &= \sum_{m=0}^{M} (\dim K_x)^m. \end{aligned}$$

These follow from choosing orthonormal bases: disjoint union gives direct-sum bases, ordered pairs give tensor-product bases, and truncation is a finite sum.

Sector constructors return projection witnesses supplied by representation data. For finite G, a unitary character $\chi : G \to U(1)$, and representation $U_E : G \to U(H_E)$, set

$$P_\chi = |G|^{-1} \sum_{g\in G} \overline{\chi(g)}\, U_E(g), \qquad H_{\mathrm{Sector}_{G,\chi}(E)} = \mathrm{Ran}(P_\chi).$$

The quotient relation is generated by $U_E(g)v - \chi(g)v$. The invariant sector is $\chi = 1$. For exchange sectors on $K^{\widehat{\otimes} n}$, $G = S_n$ and $U_E = \Pi_n$; the trivial and sign characters give

$$P_{\mathrm{s},n} = n!^{-1} \sum_{\sigma\in S_n} \Pi_n(\sigma), \qquad P_{\mathrm{a},n} = n!^{-1} \sum_{\sigma\in S_n} \mathrm{sgn}(\sigma)\Pi_n(\sigma).$$

The observable algebra is never inferred from these formulas. It is the policy slot Π_{obs}: carrier-only, full, block, tensor-local, invariant, or compressed.

9.5 Example 1: A Maybe Qubit

Let $Q = \mathbb{C}^2$ with basis e_0, e_1. Compile

$$E_1 = \mathrm{OR}(\mathbf{1}, Q).$$

The rule gives

$$H_{E_1} = \mathbb{C} \oplus \mathbb{C}^2, \qquad H_{E_1,0} = \mathbb{C}, \quad H_{E_1,1} = \mathbb{C}^2, \quad \dim H_{E_1} = 3.$$

With full observable policy, $\mathcal{A}_{E_1} = \mathcal{B}(\mathbb{C} \oplus \mathbb{C}^2)$. With block policy,

$$\mathcal{A}_{E_1}^{\mathrm{block}} = \mathcal{B}(\mathbb{C}) \oplus \mathcal{B}(\mathbb{C}^2).$$

Thus the same carrier has two different compiled observable outputs.

9.6 Example 2: Two Distinguishable Qubits

Compile $E_2 = \mathrm{AND}(Q, Q)$. The carrier is

$$H_{E_2} = Q \otimes Q, \qquad \{e_0 \otimes e_0, e_0 \otimes e_1, e_1 \otimes e_0, e_1 \otimes e_1\} \text{ is a basis,}$$

so $\dim H_{E_2} = 4$, all in degree 2. The tensor-local policy is the subalgebra generated by

$$\mathcal{B}(Q) \otimes I, \qquad I \otimes \mathcal{B}(Q),$$

whereas the full policy is $\mathcal{B}(Q \otimes Q)$. The compiler records which one was requested; it does not identify them by syntax alone.

9.7 Example 3: Derived Tensor-Power Closure

Compile the macro-expanded expression

$$E_3 = \mathsf{T}_{\leq 2}(Q) = \mathrm{OR}(\mathbf{1}, \mathrm{OR}(Q, \mathrm{AND}(Q, Q))).$$

The output carrier is

$$H_{E_3} = \mathbb{C} \oplus Q \oplus (Q \otimes Q),$$

with sector dimensions

$$\dim H_0 = 1, \qquad \dim H_1 = 2, \qquad \dim H_2 = 4, \qquad \dim H_{E_3} = 1 + 2 + 4 = 7.$$

If $U : G \to U(Q)$ is supplied, symmetry propagation gives

$$U_{E_3}(g) = 1_{\mathbb{C}} \oplus U(g) \oplus (U(g) \otimes U(g)).$$

No Bose/Fermi relation has been imposed yet; this is still the distinguishable graded representation.

9.8 Example 4: Two-Qubit Exchange-Irrep Sectors

Let S be the swap on $Q \otimes Q$. Since

$$S(e_0e_0) = e_0e_0, \quad S(e_1e_1) = e_1e_1, \quad S(e_0e_1) = e_1e_0, \quad S(e_1e_0) = e_0e_1,$$

where e_ie_j abbreviates $e_i \otimes e_j$, the exchange representation of S_2 has two one-dimensional irrep characters, trivial and sign, supplying projectors with ranges:

$$P_{\mathrm{s}} = \frac{1}{2}(I + S), \qquad P_{\mathrm{a}} = \frac{1}{2}(I - S)$$

$$\mathrm{Ran}(P_{\mathrm{s}}) = \mathrm{span}\{e_0e_0,\ (e_0e_1 + e_1e_0)/\sqrt{2},\ e_1e_1\},$$
$$\mathrm{Ran}(P_{\mathrm{a}}) = \mathrm{span}\{(e_0e_1 - e_1e_0)/\sqrt{2}\}.$$

Thus the compiler reports dimensions 3 and 1 for the trivial- and sign-irrep sectors. The Julia test suite checks the projection, orthogonality, trace/dimension, and basis-vector claims for this example.

9.9 Example 5: A Finite Invariant Quotient

Let $C_2 = \{1, s\}$ act on $H = \mathbb{C}^2$ by the swap matrix. Then

$$P_{C_2} = \frac{1}{2}(I + S) = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{pmatrix}.$$

Its range is $\mathrm{span}\{(1,1)\}$, and its kernel is $\mathrm{span}\{(1,-1)\}$. The quotient presentation is therefore

$$\mathbb{C}^2/\,\mathrm{span}\{(1,-1)\} \cong \mathrm{span}\{(1,1)\}.$$

This is $\mathrm{Sector}_{C_2,1}(H)$, the trivial-irrep sector of the C_2 representation. The invariant observable algebra on the original carrier consists of the operators commuting with S, while the compressed algebra on the invariant line is one-dimensional. These are different observable policies even in this small example.

9.10 Example 6: A Required Compile Error

Let A, B be atoms with projective symmetries over the same physical group G, but represented in the input by unrelated central extensions $E_A \to G$ and $E_B \to G$. The carrier of $\mathrm{OR}(A, B)$ is still $K_A \oplus K_B$. However, a request for one compiled symmetry representation must fail:

$$\Gamma; \Pi \vdash \mathrm{OR}(A, B) \Uparrow \text{“no common symmetry group or central extension specified”}.$$

This is not a missing feature; it is the strict semantics required by CA-05 and CONVENTIONS.md (f).

9.11 Lamport Dependency Skeleton

$D24$: $\Gamma; \Pi \vdash E \Downarrow (H_E, \{H_{E,n}\}, \mathcal{A}_E, U_E, P_E, Q_E, \mathsf{Ev}_E)$ or $\Gamma; \Pi \vdash E \Uparrow \mathsf{reason}$.
$D25$: $E ::= 0 \mid \mathbf{1} \mid x \mid \mathrm{OR} \mid \mathrm{AND} \mid \mathrm{Sector}_{G,\chi}$.
$D26$: $\mathsf{T}_{\le M}(x) = \mathrm{OR}_{m=0}^{M} x^{[m]}, \quad H_{\mathsf{T}_{\le M}} = \bigoplus_{m=0}^{M} K_x^{\otimes m}$.
$D27$: $P_\chi = |G|^{-1} \sum_g \overline{\chi(g)} U_E(g), \quad H_{\mathrm{Sector}_{G,\chi}} = \mathrm{Ran}(P_\chi)$.
$T15$: Finite dimensions obey sum, product, and $\sum_{m=0}^{M} d^m$ by explicit basis constructions.
$T16$: Irrep/character choices supply projection/subspace/quotient witnesses.
$T17$: Compiler failure is a specified outcome when requested structure is ill typed or unsupported.

9.12 Acceptance Criteria and Boundary

Future implementation work must preserve these acceptance criteria: every constructor fills typed output slots; every nontrivial mathematical output has an evidence tag; every quotient names its relation subspace; every observable claim names its observable policy; and every executable example asserts an invariant, not merely successful execution.

This compiler does not yet parse fusion-category simples, anyonic braid data, Hamiltonians, lattice limits, conformal nets, or VOA data. Those enter only after their source conventions and invariants are fixed. CA-08 is the finite Hilbert-space kernel that later shards must extend without weakening the output record, the evidence discipline, or the failure semantics.

10 Fibonacci Kinematics as a Fusion-Rule Sector

10.1 Status, Sources, and Dependencies

Status: checked kinematic shard. This shard confirms the side conjecture: before braiding, Hamiltonians, or continuum limits, the Fibonacci fusion-path Hilbert spaces are a sector construction from one physical atom τ. The vacuum $\mathbf{1}$ is still present as the tensor unit and as a charge label; it is not a second physical particle atom.

Dependencies: CA-02 for AND/OR/vacuum, CA-06 for projection/subspace semantics, CA-08 for compiler output slots, and CONVENTIONS.md (c) for the left-associated kinematic path basis.

10.2 One Physical Atom, Two Charge Labels

The physical input word is n copies of one anyon type:

$$\tau, \tau, \ldots, \tau.$$

The charge labels used to describe intermediate fusion outcomes are

$$L = \{\mathbf{1}, \tau\}.$$

Thus the proposed slogan is correct only with this distinction:

$$\text{one physical atom } \tau \quad + \quad \text{charge labels } \{\mathbf{1}, \tau\} \quad + \quad \text{fusion-rule sector selection.}$$

The vacuum $\mathbf{1}$ is required because it is both the tensor unit and a possible total charge. It is not an additional non-vacuum particle species.

10.3 Ambient Path Space and Admissibility Projector

Use the left-associated path basis fixed in CONVENTIONS.md (c). For n physical τ-atoms and total charge $c \in \{\mathbf{1}, \tau\}$, define the ambient word space

$$W_{n,c} = \ell^2\{(x_0, \ldots, x_n) : x_i \in \{\mathbf{1}, \tau\},\ x_0 = \mathbf{1},\ x_n = c\}.$$

The basis vector $|x_0, \ldots, x_n\rangle$ records the cumulative charge after successively fusing the first i copies of τ.

The Fibonacci fusion rules needed here are

$$\mathbf{1} \otimes \tau = \tau, \qquad \tau \otimes \mathbf{1} = \tau, \qquad \tau \otimes \tau = \mathbf{1} \oplus \tau.$$

Let $N_{ab}^c \in \{0, 1\}$ be these multiplicity-free fusion coefficients. Define the diagonal operator

$$P_{n,c}|x_0, \ldots, x_n\rangle = \left(\prod_{i=1}^{n} N_{x_{i-1},\tau}^{x_i}\right)|x_0, \ldots, x_n\rangle.$$

Since every diagonal entry is 0 or 1, $P_{n,c} = P_{n,c}^* = P_{n,c}^2$. The kinematic Fibonacci Hilbert space is therefore the sector

$$H_{n,c}^{\mathrm{Fib}} := \mathrm{Ran}(P_{n,c}) \subseteq W_{n,c}.$$

This is exactly the CA-06 projection/subspace semantics, but the projector is not supplied by a group character. It is supplied by categorical fusion-rule admissibility.

10.4 Dimension Recurrence

Let

$$a_n = \dim H^{\mathrm{Fib}}_{n,\mathbf{1}}, \qquad b_n = \dim H^{\mathrm{Fib}}_{n,\tau}.$$

At $n = 0$, no physical τ-atom has been fused, so

$$a_0 = 1, \qquad b_0 = 0.$$

Adding one more τ gives the transition equations

$$a_{n+1} = b_n, \qquad b_{n+1} = a_n + b_n.$$

The first equation says total charge $\mathbf{1}$ can only come from $\tau \otimes \tau \to \mathbf{1}$. The second says total charge τ can come from $\mathbf{1} \otimes \tau \to \tau$ or from $\tau \otimes \tau \to \tau$. Hence

$$b_{n+1} = b_n + b_{n-1}, \qquad b_0 = 0, \quad b_1 = 1.$$

Thus the total-τ sector dimensions are Fibonacci numbers:

$$0, 1, 1, 2, 3, 5, 8, \ldots.$$

The Julia test suite checks the pair sequence

$$(a_n, b_n) = (1, 0), (0, 1), (1, 1), (1, 2), (2, 3), (3, 5), (5, 8), (8, 13)$$

for $0 \le n \le 7$, including the recurrence.

10.5 Small Examples

$n = 1$. For total charge τ, the ambient basis is $|\mathbf{1}, \tau\rangle$, and it is admissible. Thus $H^{\mathrm{Fib}}_{1,\tau} \cong \mathbb{C}$.

$n = 2$. The admissible paths are

$$|\mathbf{1}, \tau, \mathbf{1}\rangle \in H^{\mathrm{Fib}}_{2,\mathbf{1}}, \qquad |\mathbf{1}, \tau, \tau\rangle \in H^{\mathrm{Fib}}_{2,\tau}.$$

This is the kinematic content of $\tau \otimes \tau = \mathbf{1} \oplus \tau$.

$n = 3$. The total-τ ambient basis has two admissible paths:

$$|\mathbf{1}, \tau, \mathbf{1}, \tau\rangle, \qquad |\mathbf{1}, \tau, \tau, \tau\rangle.$$

The forbidden word $|\mathbf{1}, \mathbf{1}, \tau, \tau\rangle$, for example, is killed because $N^{\mathbf{1}}_{\mathbf{1},\tau} = 0$. Therefore $\dim H^{\mathrm{Fib}}_{3,\tau} = 2$.

10.6 Compiler Consequence

This is the first example where the Hilbert-space compiler wants category data. At the carrier level, it can compile the ambient finite word space $W_{n,c}$ and the diagonal sector projection $P_{n,c}$. But producing $P_{n,c}$ from the expression "n copies of τ" requires the fusion coefficients N^c_{ab}. That is already tensor-category input, not ordinary Hilbert-space grammar.

Moreover, the source makes clear that different fusion orders give different fusion-tree bases and are related by F-moves. Therefore a full Fibonacci compiler must eventually output not only

$$H^{\mathrm{Fib}}_{n,c}$$

but also the coherent recoupling data that identifies different bracketings: the F-symbols satisfying the pentagon equations, and later the R-symbols for braiding. The current shard is only kinematic: it builds the admissible sector and counts it.

10.7 Lamport Dependency Skeleton

$$\begin{aligned}
D28 &: L = \{\mathbf{1}, \tau\}, \quad \mathbf{1} \otimes \tau = \tau, \quad \tau \otimes \tau = \mathbf{1} \oplus \tau. \\
D29 &: W_{n,c} = \ell^2\{(x_0, \ldots, x_n) : x_0 = \mathbf{1},\ x_n = c,\ x_i \in L\}. \\
D30 &: P_{n,c}|x\rangle = \prod_i N_{x_{i-1},\tau}^{x_i} |x\rangle. \\
D31 &: H_{n,c}^{\mathrm{Fib}} = \mathrm{Ran}(P_{n,c}). \\
L5 &: P_{n,c} = P_{n,c}^* = P_{n,c}^2. \\
T18 &: \dim H_{n,\tau}^{\mathrm{Fib}} \text{ obeys } b_{n+1} = b_n + b_{n-1},\ b_0 = 0,\ b_1 = 1. \\
T19 &: \text{Full Fibonacci compilation needs } F\text{-coherent tensor-category data.}
\end{aligned}$$

10.8 Boundary

This shard does not fix the Fibonacci F-matrix gauge, any R-symbols, braid representations, golden-chain Hamiltonians, or a continuum limit. It only realises the fusion-path Hilbert spaces kinematically as admissibility sectors.

11 Why Lattice Symmetry Generators Are a Problem

11.1 Status, Sources, and Dependencies

Status: proposal shard with sourced boundary. The claim that a lattice boost should look like a first moment of the Hamiltonian density is an intuition until a local source, derivation, or test is attached. This shard sets the boundary for CA-10–CA-17.

Dependencies: CA-05 for symmetry as unitary/projective-unitary representation, CA-08 for compiler output slots, and CONVENTIONS.md (h) for lattice candidate generator signs.

11.2 The Minimal Quantum-Mechanical Input

An unadorned quantum system gives a Hilbert space and, once dynamics is chosen, a self-adjoint Hamiltonian. By Stone's theorem this is the same kind of datum as a strongly continuous unitary representation of the additive group $\mathbb{R}$:

$$U(t) = \exp(-itH).$$

Thus the one symmetry forced by the bare dynamical presentation is time translation. Everything beyond this requires extra structure: spatial coordinates, locality, a metric, a lattice graph, or a continuum target.

11.3 What the Lattice Adds

A lattice Hamiltonian is not merely a self-adjoint operator. It usually comes as a decomposition

$$H = \sum_j h_j$$

where each h_j is local near a site, bond, cell, or other geometric support. The cited lattice-fermion source uses exactly this pattern: a Hamiltonian is a sum of self-adjoint finite-support terms, together with an observable algebra whose generators are intended as discretisations of continuum fields.

This decomposition is extra data. Two different decompositions of the same operator H can lead to different candidate densities and hence different candidate symmetry generators. Therefore the compiler target is not

$$\text{Hilbert space plus one operator } H,$$

but rather

$$(\text{Hilbert space, lattice geometry, } \{h_j\}, \text{ scaling map}).$$

11.4 The Continuum Target

If the intended continuum theory is Lorentz-invariant in $d+1$ dimensions, the target symmetry is not just $\mathbb{R}$-time translation. It is a projective-unitary representation of the Poincare group, lifted as necessary to a unitary representation of a covering group. Its translation subgroup has self-adjoint generators

$$P_0 = H, \qquad P_1, \ldots, P_d,$$

where the P_a are momenta.

Motivating question. If the microscopic input only supplies a lattice Hamiltonian density $H = \sum_j h_j$, how do we guess the lattice analogues of

$$P_a, \qquad M_{ab} \text{ rotations}, \qquad M_{0a} \text{ boosts?}$$

11.5 Intuition to Be Checked

Intuition 1: a rotation is a position-dependent spatial translation. In the plane, the vector field for an infinitesimal rotation about the origin is

$$v(x, y) = (-y, x).$$

Thus if we can identify local momentum densities, rotations should be spatial first moments of those densities.

Intuition 2: a boost is a position-dependent time translation. Since time translation is generated by energy, the one-dimensional boost candidate is a first moment of the energy density,

$$K_x \propto \sum_j x_j h_j.$$

Intuition 3: the commutator of a boost with time translation should produce spatial translation, up to sign and normalization. On the lattice this suggests that

$$i[H, K_x]$$

is a candidate momentum generator. CA-12 derives the nearest-neighbour formula from this ansatz and records the sign convention explicitly.

11.6 Source Boundary

Weinberg QFT Vol. 1 is now registered with targeted OCR anchors, and the original Koo–Saleur source is now registered from arXiv with source-TeX and PDF text anchors. The present shard still cites Schottenloher for continuum symmetry definitions, OAR/wavelet sources for scaling-limit translation recovery, and the local lattice-fermion source for rigorous free-fermion Koo–Saleur convergence. CA-14 cites the original Koo–Saleur paper for the source-local lattice Virasoro prototype.

12 The Continuum Bulk Symmetry Target

12.1 Status, Sources, and Dependencies

Status: sourced definitions plus checked local derivation. The Poincare group and the projective-unitary implementation are sourced. The commutator table below is a local vector-field derivation from the sourced metric-preserving description, and its signs are pinned by a Julia check.

Dependencies: CA-05, CA-10, and CONVENTIONS.md (h), (k).

12.2 Group-Level Target

For a $D = d + 1$ dimensional Lorentzian target, Schottenloher describes the Poincare group as the semidirect product of the Lorentz group and translations:

$$P = SO_0(1, d) \ltimes \mathbb{R}^{d+1},$$

with the usual replacement by the relevant covering group when the quantum symmetry is implemented projectively. The translation subgroup is abelian and has commuting self-adjoint generators P_μ. In the source convention,

$$P_0 = H, \qquad P_a \text{ are spatial momenta } (a = 1, \dots, d).$$

For Euclidean two-dimensional CFT language, the Euclidean motion group is generated by rotations and translations. This gives the Wick-rotated analogue: rotations and translations are the geometric target before scale/conformal extensions are added.

12.3 Vector-Field Derivation

Let $\eta = \mathrm{diag}(1, -1, \dots, -1)$, and lower indices by $x_\mu = \eta_{\mu\nu} x^\nu$. Use vector fields

$$P_\mu = \partial_\mu, \qquad M_{\mu\nu} = x_\mu \partial_\nu - x_\nu \partial_\mu, \qquad M_{\mu\nu} = -M_{\nu\mu}.$$

The source gives the metric-preserving Lie algebra condition $\omega^T \eta + \eta\omega = 0$. The $M_{\mu\nu}$ form the corresponding coordinate vector fields, while P_μ are translations.

Compute the Lie bracket of vector fields. Since the P_μ have constant coefficients,

$$[P_\mu, P_\nu] = 0.$$

For the mixed bracket,

$$\begin{aligned}[M_{\mu\nu}, P_\rho] &= -\partial_\rho(x_\mu)\partial_\nu + \partial_\rho(x_\nu)\partial_\mu \\ &= \eta_{\nu\rho} P_\mu - \eta_{\mu\rho} P_\nu.\end{aligned}$$

For Lorentz-Lorentz brackets,

$$\begin{aligned}[M_{\mu\nu}, M_{\rho\sigma}] = \eta_{\nu\rho} M_{\mu\sigma} &- \eta_{\mu\rho} M_{\nu\sigma} \\ &+ \eta_{\mu\sigma} M_{\nu\rho} - \eta_{\nu\sigma} M_{\mu\rho}.\end{aligned}$$

The Julia check evaluates representative brackets using this convention. For example in $3 + 1$ dimensions,

$$[M_{01}, P_0] = -P_1, \qquad [M_{01}, P_1] = -P_0.$$

12.4 Boost-Time Relation

With $K_a := M_{0a}$, the vector-field convention gives

$$[K_a, P_0] = -P_a.$$

Equivalently, $[P_0, K_a] = P_a$. The passage from this vector-field statement to self-adjoint commutator notation is the separate sign convention in CONVENTIONS.md (k): Stone generators use $U_X(t) = \exp(-itA_X)$, the spacetime flow is implemented passively, and the resulting target is

$$i[H, K_a] = P_a, \qquad i[P_b, K_a] = \delta_{ab} H.$$

Spatial rotations carry the additional convention $J_{ab} = -A_{M_{ab}}$, which is why the Gaussian one-particle table has $i[K_a, K_b] = J_{ab}$ even though the active vector fields obey $[M_{0a}, M_{0b}] = -M_{ab}$. CA-12 keeps finite-lattice signs explicit under CONVENTIONS.md (h).

12.5 Boundary

This shard does not claim that a finite lattice has an exact Poincare representation. It only identifies the continuum algebraic target whose relations a sequence of lattice candidates should approximate after scaling.

13 A One-Dimensional Boost Produces a Momentum Current

13.1 Status, Sources, and Dependencies

Status: checked local derivation. The identity in this shard is not quoted from a source. It is derived from the local Hamiltonian-density ansatz and checked by a small coefficient invariant.

Dependencies: CA-10, CA-11, and CONVENTIONS.md (h).

13.2 Setup

Let a finite open one-dimensional lattice have sites or bonds labelled $j = 1, \dots, N$, real positions x_j, and a self-adjoint local-density decomposition

$$H = \sum_{j=1}^{N} h_j.$$

Assume the density terms commute at distance greater than one:

$$[h_j, h_k] = 0 \qquad \text{if } |j - k| > 1.$$

This is the finite-range simplification appropriate for the first derivation; longer-range densities only add longer-range edge terms.

Define the position-weighted energy moment

$$K_x = \sum_{j=1}^{N} x_j h_j.$$

This is the lattice boost candidate in the sense of CA-10: a boost is treated as a position-dependent time translation, and time translation is generated by energy density.

13.3 Commutator Calculation

Compute

$$\begin{aligned} i[H, K_x] &= i \left[\sum_j h_j, \sum_k x_k h_k \right] \\ &= i \sum_{j,k} x_k [h_j, h_k]. \end{aligned}$$

The terms with $j = k$ vanish, and by the nearest-neighbour assumption only pairs $(j, j+1)$ survive. For each adjacent pair,

$$i x_{j+1} [h_j, h_{j+1}] + i x_j [h_{j+1}, h_j] = i(x_{j+1} - x_j)[h_j, h_{j+1}].$$

Hence

$$i[H, K_x] = \sum_{j=1}^{N-1} (x_{j+1} - x_j)\, i[h_j, h_{j+1}].$$

Each $i[h_j, h_{j+1}]$ is self-adjoint because h_j, h_{j+1} are self-adjoint.

13.4 Uniform Spacing

If $x_j = aj$, then $x_{j+1} - x_j = a$ and

$$i[H, K_x] = a \sum_{j=1}^{N-1} i[h_j, h_{j+1}].$$

For unit spacing this gives the candidate bulk momentum

$$P_{\text{bulk}}^{\text{lat}} := \sum_{j=1}^{N-1} i[h_j, h_{j+1}],$$

with endpoint terms deliberately visible. Periodic chains require an extra boundary convention because the position function j is not single-valued on a circle.

13.5 Interpretation

The continuum bracket in CA-11 says that boost plus time translation gives a spatial translation, up to sign convention. The lattice calculation says that the same operation applied to the first energy moment produces the commutator of neighbouring energy densities. Therefore $i[h_j, h_{j+1}]$ is not guessed from thin air: it is the local current forced by the first-moment ansatz.

13.6 Boundary

This is not yet a theorem that $P_{\text{bulk}}^{\text{lat}}$ converges to continuum momentum. The missing ingredients are a scaling map, a domain for the candidate operators, a state/sector sequence, and convergence of the commutator relations. CA-15 turns these into acceptance checks.

14 Position-Dependent Bulk Generators

14.1 Status, Sources, and Dependencies

Status: proposal shard with local derivation pattern. The sourced continuum content is geometric: vector fields, translations, and rotations. The higher-dimensional lattice formulas are compiler targets, not established convergence claims, and this shard does not define a stress-energy tensor or a lattice momentum density.

Dependencies: CA-10–CA-12 and CONVENTIONS.md (h).

14.2 Continuum Picture

Translations are flows of constant vector fields. In the Euclidean plane, the infinitesimal motion

$$X(z) = c + i\theta z$$

contains both translations (c) and rotations ($i\theta z$). Written in real coordinates, the rotation part is the position-dependent vector field

$$(x, y) \mapsto (-\theta y, \theta x).$$

This is the precise continuum content of the slogan:

$$\text{rotation} = \text{position-dependent translation.}$$

14.3 Scalar Ramps

Suppose lattice density terms are attached to positions $r \in \Lambda \subset \mathbb{R}^d$. For a scalar function $f : \Lambda \to \mathbb{R}$, define

$$K_f := \sum_{r\in\Lambda} f(r)h_r.$$

The same derivation as CA-12 gives, for commuting non-overlapping densities,

$$i[H, K_f] = \sum_{\{r,s\}} (f(s) - f(r))\, i[h_r, h_s],$$

where the sum is over unordered overlapping pairs after a choice of orientation. This is a discrete-gradient response. Linear ramps

$$f_a(r) = r^a$$

are therefore the first candidates for spatial momentum components.

14.4 Boost Candidates

Boosts can be attempted directly from energy density:

$$K_a^{\text{lat}} := \sum_{r\in\Lambda} r^a h_r.$$

Then $i[H, K_a^{\text{lat}}]$ gives the a-direction current candidate. This is the higher-dimensional version of the one-dimensional derivation, but it depends on the chosen density supports and orientations.

14.5 Rotation Candidates

Rotations require one more layer. In continuum notation,

$$J_{ab} = \int \left(x_a p_b(x) - x_b p_a(x) \right) dx.$$

Here this formula is only a continuum mnemonic for why rotations should involve an angular-momentum density. The Schottenloher anchors above support the underlying source vector fields and rotations, not a QFT stress-energy tensor, not a momentum-density formula, and not a lattice dictionary for $p_a(x)$. Thus an actual lattice rotation candidate must be routed through a later momentum-density/current convention rather than built directly from h_r. Once such a convention is fixed, the provisional compiler shape is:

$$J_{ab}^{\mathrm{lat}} \approx \sum_r \big(r_a p_b^{\mathrm{lat}}(r) - r_b p_a^{\mathrm{lat}}(r) \big),$$

where p_a^{lat} must come with its own source, derivation, or checked continuity-equation evidence.

14.6 Compiler Moral

The lattice Hilbert-space compiler cannot infer these operators from the carrier alone. It needs a lattice geometry, a local Hamiltonian-density decomposition, and a target vector-field algebra. The output is not an exact symmetry by default; it is a candidate list with proof obligations.

14.7 Boundary

No stress-energy convention is fixed here. No cell-orientation convention is fixed here. No periodic wrapping convention is fixed here. No higher-dimensional convergence theorem is fixed here. Those choices must be made in a later shard before any numerical or categorical lattice model uses these candidates as checked symmetries.

15 Koo–Saleur as the Prototype

15.1 Status, Sources, and Dependencies

Status: source-local prototype plus later rigorous precedent. The original Koo–Saleur paper is registered here and supports the lattice Virasoro prototype as a conjectural construction with numerical and analytic checks for XXZ/RSOS/Temperley–Lieb settings. This shard keeps that separate from the local lattice-fermion/OAR treatment, which states and proves a free-fermion Koo–Saleur convergence result.

Dependencies: CA-10–CA-13.

15.2 Continuum Side

In a two-dimensional CFT on the circle, the Hamiltonian is expressed as an integral of a Hamiltonian density,

$$H = \int h(x)\, dx.$$

The source identifies $h(x)$ as a local quantum field built from the chiral and anti-chiral energy-momentum tensors $T, \bar{T}$. Its Fourier modes H_k are linear combinations of the Virasoro generators $L_k, \bar{L}_k$.

Thus in this special setting the desired spacetime symmetries are not abstract: they are generated by modes of the stress tensor.

15.3 Lattice Side

The Koo–Saleur move is to imitate the continuum construction using the lattice Hamiltonian density. From a lattice density

$$h^{(N)} : \Lambda_N \to \mathcal{A}_N$$

one forms Fourier modes $H_k^{(N)}$. In the original Temperley–Lieb presentation, the Hamiltonian-limit density is represented by ground-state-subtracted $e_j - e_\infty$ terms, while the second stress-tensor component is represented by nearest-neighbour commutators $[e_j, e_{j+1}]$. The lattice Virasoro candidates $l_n, \bar{l}_{-n}$ are therefore Fourier sums of

$$(e_j - e_\infty) \;\pm\; \text{constant} \cdot [e_j, e_{j+1}],$$

with the $c/24$ zero-mode shift. The later free-fermion/OAR Definition 4.1 uses the same structural pattern: density modes plus commutator corrections against the lattice Hamiltonian mode $H_0^{(N)}$ produce lattice analogues of L_k and $\bar{L}_k$.

The important structural point for our programme is:

$$\text{lattice symmetry generator} = \text{weighted Hamiltonian density} + \text{commutator correction}.$$

This is the same motif as CA-12, but now with Fourier weights and a Virasoro target rather than a first moment and a Poincare translation target.

15.4 Scaling Representation Matters

The original source already builds in ground-state subtraction, normal ordering, and a double-limit process; it also warns that finite-L lattice quantities do not close as the Virasoro algebra and that a scaling limit of a commutator is not generally the commutator of scaling limits. The later rigorous free-fermion/OAR source similarly makes convergence depend on the correct scaling-limit representation and on subtracting ground-state expectation values, corresponding to normal ordering. Therefore the compiler must not output only formal sums. It must also output:

state sequence,
renormalisation/scaling maps,
vacuum subtraction policy,
operator topology target.

15.5 What This Foreshadows

Koo–Saleur is a prototype for the lattice-symmetry compiler:

$$\{h_j\} \longmapsto \{H_k^{(N)}, L_k^{(N)}, \bar{L}_k^{(N)}\} \longrightarrow \{L_k, \bar{L}_k\}.$$

The first arrow is algebraic and finite-scale. The second is analytic and requires a scaling theorem.

15.6 Boundary

The original Koo–Saleur source supports the prototype as a conjectural and numerically checked construction for XXZ/RSOS/Temperley–Lieb lattice models; it does not prove a general rigorous convergence theorem for arbitrary microscopic models. The convergence theorem cited here is still the later free-fermion/OAR result, and arbitrary anyon chains remain outside both sourced claims.

16 Acceptance Tests for Lattice Symmetry Candidates

16.1 Status, Sources, and Dependencies

Status: specification shard. This shard does not prove convergence for new models. It states the checks required before a candidate generator is promoted from proposal to checked continuum symmetry data.

Dependencies: CA-10–CA-14.

16.2 Local Finite-Scale Checks

At fixed lattice size N, a candidate generator G_N must first pass finite operator checks:

$$G_N = G_N^*, \qquad G_N \text{ is built from named local density terms}, \qquad \mathrm{supp}(G_N) \text{ policy is explicit.}$$

For unbounded thermodynamic-limit candidates, the finite check is only the first layer; domains must later be named.

16.3 Algebraic Checks

Given a target Lie algebra with generators X_a, the lattice candidates $G_a^{(N)}$ should satisfy commutator closure in a controlled sense:

$$i[G_a^{(N)}, G_b^{(N)}] \approx f_{ab}{}^{c} G_c^{(N)} + \text{central or boundary terms.}$$

The approximation relation must be specified: exact finite-size equality, matrix-element convergence, strong resolvent convergence, strong operator convergence on a core, or convergence after compression to a low-energy sector.

16.4 Scaling Checks

The OAR source shows that continuous spatial translations can be reconstructed from discrete dyadic translations in a suitable scaling limit, with the usual momentum operators as generators. It also states that analogous convergence for Lorentz or conformal transformations requires further work. This is the main warning for this programme:

$$\text{translation recovery} \not\Rightarrow \text{Poincare recovery.}$$

Therefore every candidate boost or rotation needs its own scaling statement. For a sequence G_N, the report must say:

$$G_N \xrightarrow{?} G$$

and name the topology, state sequence, renormalisation maps, and dense domain or core on which the statement is intended.

16.5 Observable Covariance Checks

The target is not only an operator on states. It is a symmetry action on the observable system. The evidence payload should include

$$\alpha_N^G(A) = e^{itG_N} A e^{-itG_N}$$

or the infinitesimal commutator $i[G_N, A]$, together with the expected transformation of local algebras. For translations this means moving supports. For rotations it means rotating supports. For boosts it means mixing time and space in the continuum representation.

16.6 Acceptance Payload

A lattice generator candidate is accepted into the compiler only with:

1. density provenance: the exact h_j and supports,
2. finite algebra checks: self-adjointness and commutators,
3. renormalisation maps and state sequence,
4. operator topology or matrix-element convergence target,
5. observable covariance statement,
6. boundary/central/vacuum-subtraction policy.

16.7 Boundary

Time-zero net axioms and Lieb-Robinson bounds give important locality control, but local causality is not the same as full Poincare covariance. The compiler must keep these evidence levels separate.

17 A Compiler Interface for Lattice Symmetry Candidates

17.1 Status, Sources, and Dependencies

Status: compiler specification shard. This shard extends CA-08 at the level of output slots. It is not yet an implementation.

Dependencies: CA-08 and CA-10–CA-15.

17.2 Input Contract

The lattice symmetry compiler receives a compiled Hilbert/QM system plus:

$$\begin{array}{ll}
\Lambda_N & : \text{finite lattice, graph, or cell complex,} \\
x & : \Lambda_N \to \mathbb{R}^d \text{ position map,} \\
\mathcal{A}_N & : \text{observable algebra with locality map,} \\
H_N = \sum_r h_r & : \text{self-adjoint local Hamiltonian density,} \\
T & : \text{target symmetry algebra or group,} \\
S_N & : \text{state/sector sequence and scaling maps, if known.}
\end{array}$$

If the Hamiltonian is supplied only as a black-box matrix, the compiler must fail for boost/rotation generation: the density decomposition is part of the mathematical data.

17.3 Candidate Generation

For one-dimensional open chains, the first candidate layer is deterministic:

$$K_x = \sum_j x_j h_j, \qquad P_x = i[H, K_x].$$

For uniform nearest-neighbour density this reduces to the CA-12 current sum.

For higher-dimensional lattices, scalar ramps $f_a(r) = r^a$ generate candidate momentum components

$$P_a^{\mathrm{lat}} := i[H, K_{f_a}].$$

Rotation candidates are then built only after the momentum-density/current layer has its own convention and evidence:

$$J_{ab}^{\mathrm{lat}} \approx \sum_r (r_a p_b^{\mathrm{lat}}(r) - r_b p_a^{\mathrm{lat}}(r)).$$

Until that layer is fixed, this is a deferred proposal slot rather than a formula emitted by the compiler. Koo–Saleur-style modes use Fourier weights instead of polynomial weights and include the commutator corrections specified by the cited Definition 4.1.

17.4 Output Contract

The compiler output for each candidate generator must include:

name	: for example $P_a, K_a, J_{ab}, L_k, \bar{L}_k$,
formula	: finite expression in h_r and commutators,
domain policy	: finite matrix, core, or unbounded deferral,
locality support	: site/bond/cell support accounting,
status	: Proposal, Source-local, or Checked,
evidence	: source lines, derivation, test, or run artifact,
continuum target	: target generator and expected bracket.

17.5 Failure Modes

The compiler must fail loudly when:

1. H has no density decomposition,
2. Λ_N has no position map but a boost/rotation is requested,
3. periodic boundary positions are requested without a branch convention,
4. the target algebra requires a source/convention not yet fixed,
5. a convergence claim is requested without scaling maps or states.

17.6 Compiler Principle

The finite expression and the continuum theorem are different compilation products. The first can often be generated algebraically from $(\Lambda_N, x, \{h_r\})$. The second is a proof obligation involving scaling maps, states, and operator topology. The compiler should output both slots, but it must never fill the proof slot with intuition.

18 Small Examples and the Lattice-Symmetry Research Queue

18.1 Status, Sources, and Dependencies

Status: worked examples and queue. The examples are deliberately small. They illustrate compiler behaviour, not new convergence theorems.

Dependencies: CA-10–CA-16 and CONVENTIONS.md (h), (j).

18.2 Example 1: Black-Box Quantum System

Input: a Hilbert space and a self-adjoint Hamiltonian H.

Output: time translation $U(t) = e^{-itH}$.

Failed request: infer boosts, rotations, or spatial translations. There is no lattice geometry, no position map, and no Hamiltonian density. The compiler must stop.

18.3 Example 2: Open Uniform Chain

Input:

$$H = \sum_{j=1}^{N} h_j, \qquad x_j = j, \qquad [h_j, h_k] = 0 \text{ for } |j-k| > 1.$$

Candidate boost:

$$K = \sum_j j h_j.$$

Candidate momentum:

$$P = i[H, K] = \sum_{j=1}^{N-1} i[h_j, h_{j+1}].$$

Status: checked finite derivation; no continuum convergence yet.

18.4 Example 3: Nonuniform Chain

Input: positions $x_1 < \cdots < x_N$. The same derivation gives

$$P_x = i[H, K_x] = \sum_{j=1}^{N-1} (x_{j+1} - x_j) i[h_j, h_{j+1}].$$

The coefficients are discrete spacings. The Julia test pins this invariant for uniform and nonuniform examples.

18.5 Example 4: Periodic Chain

Input: a circle with periodic density terms.

Problem: $x_j = j$ is not a single-valued function on the circle. A periodic boost-like first moment requires a branch cut, a sawtooth, Fourier modes, or a different construction. The same warning applies to finite periodic momentum-coordinate tests: a finite grid supports periodic symbols and Fourier eigenvalue checks, not an exact differential operator $X = i\partial_k$ with a canonical commutator. Koo–Saleur Fourier modes are the better sourced periodic prototype.

18.6 Example 5: Square Lattice

Input: positions $r = (m, n)$. Scalar ramps $f_x(r) = m$, $f_y(r) = n$ give two candidate momentum components via

$$P_a^{\mathrm{lat}} = i[H, \sum_r r^a h_r].$$

Rotation about the origin should then be built as a position-dependent translation from the two momentum-density components. Status: proposal until an orientation and density-localisation convention is fixed.

18.7 Example 6: Koo–Saleur Modes

Input: periodic one-dimensional critical Hamiltonian density. Fourier modes of the density and commutator corrections give candidates for $L_k, \bar{L}_k$. In the cited free-fermion/OAR setting, smeared approximants converge to smeared Virasoro generators. Status outside that setting: source-local precedent, not a general theorem.

18.8 Research Queue

1. Use registered Weinberg anchors before promoting exact Lorentz-generator formulas.
2. Use the registered original Koo–Saleur anchors before promoting model-specific KS claims.
3. Source or derive a general local energy-current theorem.
4. Fix periodic branch/sawtooth/Fourier first-moment conventions.
5. Define higher-dimensional oriented density supports.
6. Add numerical examples checking commutator closure in small chains.
7. Connect anyon-chain Hamiltonian densities to the same candidate layer.

18.9 Landing Point

The next compiler upgrade is therefore not only a Hilbert-space compiler and not yet only a tensor-category compiler. It is a lattice-QM compiler layer:

$$(\text{carrier}, \mathcal{A}_N, \Lambda_N, \{h_r\}, x, S_N) \longmapsto (\text{candidate bulk symmetries}, \text{evidence obligations}).$$

Only after those obligations are discharged can the continuum symmetry representation be claimed.

19 Why Galilean Symmetry Is a Different Lattice Target

19.1 Status, Sources, and Dependencies

Status: sourced boundary plus proposal shard. The Galilei group as a classical nonrelativistic symmetry group is sourced locally. The general projective-representation and central-extension mechanism is sourced locally. The specific mass central extension of the Galilei group is not yet sourced in this repository; CA-20 treats its coefficient as a checked canonical-commutator derivation, not as a literature claim about a named group.

Dependencies: CA-05 for projective symmetries via central extensions, CA-10–CA-17 for the Lorentzian lattice-symmetry programme, and CONVENTIONS.md (i).

19.2 The New Target

The previous block targeted Lorentz-invariant continuum systems. Its first lattice intuition was:

$$\text{Lorentz boost} \simeq \text{position-dependent time translation}, \qquad K_a^{\mathrm{Lor}} \sim \sum_x x_a h_x.$$

That is appropriate only when the continuum target mixes space and time in the Lorentzian way and the boost is controlled by the energy density.

The nonrelativistic target is different. Schottenloher lists the Galilei group as the symmetry group for free particles in classical nonrelativistic mechanics. For the lab-book compiler this means that a Galilean target is not obtained by renaming the Poincare block. It has absolute time, spatial translations, rotations, and boosts whose quantum implementation may be projective.

19.3 Immediate Consequence for Lattices

The Lorentzian boost ansatz used only the Hamiltonian density h_x. A Galilean boost needs more input. At time $t = 0$, the familiar nonrelativistic one-particle expression is controlled by the mass-weighted position, not by the energy-weighted position:

$$G_a(0) \sim mX_a.$$

For a many-body lattice this suggests a mass-density or particle-number-density first moment:

$$G_a^{\mathrm{lat}}(0) \sim \sum_{x\in\Lambda} x_a\,\mu_x, \qquad \mu_x = m\,n_x \quad \text{when the particles have common mass } m.$$

This is a proposal-level lattice formula until the model supplies the density μ_x, the conservation law, and the continuity/current relation.

19.4 Compiler Implication

The compiler rule is therefore stricter than in CA-16:

$$\begin{aligned} &\text{Lorentzian candidate layer:} && (\Lambda, x, \{h_x\}) \text{ may generate } K_a \sim \textstyle\sum_x x_a h_x, \\ &\text{Galilean candidate layer:} && (\Lambda, x, \{h_x\}) \text{ is not enough.} \end{aligned}$$

For Galilean boosts the input must also name:

$$\{\mu_x\} \text{ or } \{n_x\}, \qquad M = \sum_x \mu_x, \qquad [H, M] = 0,$$

plus, for a momentum candidate, a current witness for the discrete continuity equation. If those data are absent, the compiler must fail loudly rather than silently using the energy first moment.

19.5 Why Projective Data Matter Earlier

In the Hilbert-space grammar, projective symmetries were handled by passing to a central extension. Galilean symmetry is the first lattice target where that choice becomes operational: the mass parameter is expected to sit in the projective/central part of the quantum representation. Locally we can support only this precise statement:

projective quantum symmetries are represented by unitary representations of central extensions.

The stronger statement identifying the Galilei extension by its standard literature name remains a source-acquisition task.

19.6 Boundary for CA-18–CA-22

The follow-up block does four things:

CA-19	derives the unextended Galilei vector-field algebra,
CA-20	derives the mass central coefficient on a canonical core,
CA-21	derives the lattice first-moment continuity formula,
CA-22	turns the data into compiler rules and small examples.

No shard in this block claims that a finite lattice has exact Galilean symmetry unless the corresponding finite-dimensional representation and commutator checks are actually supplied.

20 The Galilei Algebra as Newtonian Vector Fields

20.1 Status, Sources, and Dependencies

Status: checked local derivation. The source anchor is that the Galilei group is the nonrelativistic free-particle symmetry group. The bracket table below is derived here from vector fields and checked by the Julia test suite.

Dependencies: CA-18 and CONVENTIONS.md (i).

20.2 Generators

Use Newtonian spacetime coordinates

$$(t, x_1, \dots, x_d).$$

The convention fixed in CONVENTIONS.md (i) is:

$$\begin{array}{rcll} H & = & \partial_t, & \text{time translation,} \\ P_a & = & \partial_a, & \text{spatial translation,} \\ G_a & = & t\,\partial_a, & \text{Galilean boost,} \\ J_{ab} & = & x_a\partial_b - x_b\partial_a, \quad a < b, & \text{spatial rotation.} \end{array}$$

Here $a, b, c \in \{1, \dots, d\}$. The boost vector field says that a boost changes position by a velocity times time while leaving absolute time fixed.

20.3 Translation and Boost Brackets

The translation fields have constant coefficients, hence

$$[H, P_a] = 0, \qquad [P_a, P_b] = 0.$$

Since $G_a = t\partial_a$,

$$\begin{aligned} [H, G_a] &= [\partial_t, t\partial_a] \\ &= \partial_t(t)\partial_a \\ &= P_a. \end{aligned}$$

The reverse bracket is therefore $[G_a, H] = -P_a$. Spatial translations do not differentiate the coefficient t, so

$$[P_a, G_b] = 0, \qquad [G_a, G_b] = 0.$$

This is the unextended vector-field algebra. It is not yet the projective quantum algebra; CA-20 adds the central term that is invisible to vector fields on spacetime.

20.4 Rotation on Vectors

For $J_{ab} = x_a\partial_b - x_b\partial_a$, compute against a translation:

$$\begin{aligned} [J_{ab}, P_c] &= -\partial_c(x_a)\partial_b + \partial_c(x_b)\partial_a \\ &= \delta_{bc}P_a - \delta_{ac}P_b. \end{aligned}$$

The same calculation applies to boosts because J_{ab} does not act on t:

$$[J_{ab}, G_c] = \delta_{bc}G_a - \delta_{ac}G_b.$$

Thus both P_c and G_c transform as spatial vectors.

20.5 Rotation-Rotation Bracket

For two rotations, expanding the vector-field bracket gives

$$[J_{ab}, J_{cd}] = \delta_{bc}J_{ad} - \delta_{ac}J_{bd} \\ + \delta_{ad}J_{bc} - \delta_{bd}J_{ac},$$

with the convention $J_{ba} = -J_{ab}$ and $J_{aa} = 0$. In $d = 3$, one representative checked bracket is

$$[J_{12}, J_{23}] = J_{13}.$$

20.6 Checked Representatives

The test suite pins the signs with the following representative outcomes:

$$\begin{aligned} [H, G_1] &= P_1, \\ [G_1, H] &= -P_1, \\ [J_{12}, P_1] &= -P_2, \\ [J_{12}, P_2] &= P_1, \\ [J_{12}, G_1] &= -G_2, \\ [J_{12}, J_{23}] &= J_{13}. \end{aligned}$$

It also checks that $[P_1, G_2] = 0$ and $[G_1, G_2] = 0$ in the unextended vector-field algebra.

20.7 Boundary

The signs here are signs for vector fields. A quantum model uses self-adjoint generators and unitary flows, so translating this table into operator commutators requires the Stone convention and the chosen $i[A, B]$ observable action. The report therefore cites this shard for the geometric algebra and CA-20 for the central quantum coefficient.

21 Projective Galilean Symmetry and the Mass Central Term

21.1 Status, Sources, and Dependencies

Status: sourced projective framework plus checked local derivation. Schottenloher supplies the representation-theoretic framework: projective symmetries need not lift to the original group, but lift through a central extension. The coefficient $m\delta_{ab}$ below is a local canonical commutator derivation checked in Julia. A local source specifically identifying the Galilei mass central extension by its standard name has not yet been registered.

Dependencies: CA-05, CA-18, CA-19, and CONVENTIONS.md (f), (i).

21.2 Why This Belongs in the Grammar

CA-05 made a conservative grammar choice: a projective symmetry is represented by an honest unitary representation of a named central extension

$$1 \longrightarrow U(1) \longrightarrow E \longrightarrow G \longrightarrow 1.$$

For Galilean symmetry this is not a decorative detail. The central parameter is physically interpreted as mass in the canonical nonrelativistic model, and it controls how boosts and momenta compose at the quantum level.

21.3 Canonical One-Particle Calculation

Work formally on a common invariant core where the position and momentum operators obey

$$[X_a, P_b] = i\delta_{ab}I.$$

For a particle of mass m, the time-zero boost generator is represented by

$$G_a(0) = mX_a.$$

Then

$$[G_a(0), P_b] = [mX_a, P_b]$$

$$= m[X_a, P_b]$$

$$= i\, m\, \delta_{ab} I.$$

The Julia check records exactly the real coefficient $m\delta_{ab}$.

21.4 Compatibility with Time Evolution

For the free Hamiltonian

$$H = \frac{1}{2m}\sum_b P_b^2,$$

the same canonical commutator gives

$$[H, mX_a] = \frac{1}{2}\sum_b [P_b^2, X_a]$$

$$= \frac{1}{2}\sum_b \big(P_b[P_b, X_a] + [P_b, X_a]P_b\big)$$

$$= \frac{1}{2}\big(-iP_a - iP_a\big) = -iP_a.$$

Thus

$$i[H, G_a(0)] = P_a.$$

This matches the CA-19 vector-field relation $[H, G_a] = P_a$ after applying the self-adjoint-generator convention used for observable evolution.

21.5 Central Extension Versus Vector Fields

The unextended vector-field algebra had $[G_a, P_b] = 0$. The canonical quantum calculation has instead

$$[G_a, P_b] = i\, m\, \delta_{ab} I.$$

The additional term is central: it is a scalar multiple of the identity on a fixed-mass sector. This is the algebraic reason the compiler must attach a central-extension witness to Galilean symmetry data rather than store only the spacetime vector fields.

21.6 Source Gap

The local source collection includes Bargmann's theorem about lifting projective representations under a Lie-algebra cohomology hypothesis, and it includes the general central-extension theorem for projective representations. It does not yet include a registered source for the standard mass central extension of the Galilei group. Until that source is acquired, this shard uses the cautious name *mass central term* and cites the checked canonical calculation as its local evidence.

21.7 Compiler Payload

A Galilean projective symmetry record must therefore include:

base group/algebra	Galilei target from CA-19,
central generator	M or I on a fixed-mass sector,
central charge	m or a mass operator,
raw commutator witness	$[G_a, P_b] = i\delta_{ab} M$,
sign convention	conversion to $i[A, B]$ stated explicitly.

An OR of sectors with different masses can still be a Hilbert direct sum, but the central element must then be represented as a block-diagonal operator rather than a single scalar charge.

22 Lattice Galilean Boosts from Mass-Density First Moments

22.1 Status, Sources, and Dependencies

Status: proposal plus checked local derivation. Local sources support Galilean invariance as a physical assumption in nonrelativistic fermion models and give examples where conservation laws produce continuity equations. The finite open-chain first-moment formula below is a local derivation checked by the Julia test suite.

Dependencies: CA-18–CA-20 and CONVENTIONS.md (h), (i).

22.2 Input Data

Let $\Lambda = \{1, \ldots, N\}$ be a finite open chain with positions x_j. A Galilean lattice candidate needs, in addition to the Hamiltonian, a conserved mass or particle-number density:

$$M = \sum_j \mu_j, \qquad [H, M] = 0.$$

For identical particles of mass m, a common special case is

$$\mu_j = m n_j,$$

where n_j is a number density.

The time-zero Galilean boost candidate is the first mass moment

$$G^{\text{lat}}(0) = \sum_j x_j \mu_j.$$

This is the Galilean analogue of mX, not the Lorentzian energy moment $\sum_j x_j h_j$.

22.3 Discrete Continuity Assumption

Assume a nearest-neighbour open-chain continuity equation with bond currents $J_{j+1/2}$:

$$i[H, \mu_j] = J_{j-1/2} - J_{j+1/2},$$

where endpoint currents are either absent or treated as boundary terms. This assumption is model data. It does not follow merely from writing $H = \sum_j h_j$.

22.4 First-Moment Derivation

Multiply the continuity equation by x_j and sum:

$$\begin{aligned} i\left[H, \sum_j x_j \mu_j\right] &= \sum_j x_j (J_{j-1/2} - J_{j+1/2}) \\ &= \sum_{j=1}^{N-1} (x_{j+1} - x_j) J_{j+1/2} \quad + \quad \text{boundary terms.} \end{aligned}$$

For a unit-spaced open chain away from endpoints,

$$i[H, G^{\text{lat}}(0)] \simeq \sum_{j=1}^{N-1} J_{j+1/2}.$$

This is the lattice momentum candidate associated with the Galilean boost. For nonuniform positions the coefficients are the spacings

$$\Delta x_j = x_{j+1} - x_j,$$

and the test suite records a coefficient check for this shard.

22.5 Relation to Center of Mass

The local paired-fermion source uses Galilean invariance to explain a response in terms of center-of-mass acceleration while relative motion is unaffected. That is exactly the intuition this shard turns into compiler data:

$$\text{Galilean boost} \quad \rightsquigarrow \quad \text{center-of-mass / mass-first-moment generator.}$$

The source does not prove the lattice formula above. It supports the physical role of center-of-mass motion under Galilean invariance; the lattice identity is the checked local derivation.

22.6 What Is and Is Not Proven

The finite identity proves only that, given a local continuity equation, the time derivative of the mass first moment is a current sum. It does not prove:

1. that the current sum converges to continuum momentum,
2. that the finite lattice has exact Galilei symmetry,
3. that the mass current equals momentum density in every model,
4. that boundary terms vanish without a stated boundary condition.

Those are acceptance-test obligations for later shards or model-specific examples.

23 Compiler Rules and Examples for Galilean Targets

23.1 Status, Sources, and Dependencies

Status: compiler specification shard. This shard does not implement the compiler. It specifies the extra fields a Galilean target requires, using the checked algebra and first-moment derivations from CA-19–CA-21.

Dependencies: CA-08, CA-16, and CA-18–CA-21.

23.2 Target Declaration

A Galilean compiler target is a structured request:

$$\mathrm{Target}(\mathrm{Gal}(d),\ \mathrm{projective},\ \mathrm{mass}).$$

It asks for:

H	time translation,
P_a	spatial translations / momenta,
J_{ab}	rotations,
G_a	Galilean boosts,
M	central mass generator.

The finite candidate layer is allowed to output formulas and witnesses. It is not allowed to assert continuum Galilean covariance without scaling-limit evidence.

23.3 Required Input Slots

Compared with CA-16, the Galilean target adds the following required slots:

μ_x or n_x	local mass or particle-number density,
$M = \sum_x \mu_x$	conserved total mass/number,
$[H, M] = 0$	finite commutator witness,
$J_{x,e}$	current witness for a discrete continuity equation,
m or M	central charge/operator for the projective representation,
boundary policy	open, periodic with branch, or thermodynamic-limit handling.

If any of the first four are absent, boosts and momenta for a Galilean target are underdetermined.

23.4 Generated Candidates

For an open chain, the deterministic candidate formulas are

$$G^{\mathrm{lat}}(0) = \sum_j x_j \mu_j,$$

and, when a continuity witness is supplied,

$$P^{\mathrm{lat}} := i[H, G^{\mathrm{lat}}(0)] = \sum_{j=1}^{N-1} (x_{j+1} - x_j) J_{j+1/2} \quad + \quad \text{boundary terms.}$$

Rotations in higher dimension require momentum or current-density data:

$$J_{ab}^{\mathrm{lat}} \approx \sum_x (x_a p_b(x) - x_b p_a(x)).$$

Thus rotations are second-stage candidates: first compile current/momentum density, then form the spatial first moment.

23.5 Example 1: One Free Particle

Input:

$$\mathcal{H} = L^2(\mathbb{R}^d), \qquad H = \frac{P^2}{2m}, \qquad [X_a, P_b] = i\delta_{ab} I.$$

Output at $t = 0$:

$$G_a = mX_a, \qquad [G_a, P_b] = im\delta_{ab} I, \qquad i[H, G_a] = P_a.$$

Status: checked formal derivation on a common core, with unbounded-operator domain work deferred.

23.6 Example 2: Number-Conserving Open Chain

Input:

$$H, \quad n_j, \quad [H, \sum_j n_j] = 0, \quad i[H, n_j] = J_{j-1/2} - J_{j+1/2}.$$

For identical mass m,

$$\mu_j = mn_j, \qquad G^{\mathrm{lat}}(0) = m \sum_j x_j n_j.$$

The compiler may output the current-sum momentum candidate with coefficient $m(x_{j+1} - x_j)$. It must still mark continuum Galilean covariance as a proof obligation.

23.7 Example 3: Hamiltonian Density Only

Input:

$$H = \sum_j h_j$$

with no n_j, no μ_j, and no current witness. For a Lorentzian target, CA-12 permits the energy-momentum proposal. For a Galilean target, the compiler must return:

```
CompileError: missing mass/number density and continuity witness.
```

Using $\sum_j x_j h_j$ here would be a convention bug.

23.8 Example 4: Spin Chain with No Particle Number

A spin chain may have a perfectly good Hamiltonian density and exact lattice translations. Unless a conserved density is explicitly declared as mass or particle number, it does not compile as a Galilean particle system. The correct output is either a failed Galilean target or a different target declaration.

23.9 Example 5: Direct Sum of Mass Sectors

For

$$\mathcal{H} = \mathcal{H}_{m_1} \oplus \mathcal{H}_{m_2},$$

the central generator is not one scalar if $m_1 \neq m_2$. It is

$$M = m_1 I_{\mathcal{H}_{m_1}} \oplus m_2 I_{\mathcal{H}_{m_2}}.$$

The compiler can carry the OR carrier, but an irreducible fixed-mass sector selection is a separate projection/sector step in the sense of CA-06.

23.10 Next Obligations

The next Galilean work needs:

1. a registered source for the standard mass central extension,
2. model examples with explicit lattice number-current continuity,
3. finite commutator tests for those models,
4. scaling maps proving that current sums converge to continuum momenta,
5. a decision on periodic boundary branch conventions.

Only after these are present should the compiler promote Galilean lattice candidates from formulas to continuum symmetry evidence.

24 Gaussian Boson Lorentz Generator Roadmap

24.1 Status, Sources, and Dependencies

Status: roadmap shard with checked seed. This shard scopes a multi-shard programme. CA-24 supplies the first checked symbol calculation. The stronger continuum conclusion is not yet proved for a general coefficient class.

Dependencies: CA-10–CA-17 for lattice Lorentz motivation and acceptance tests, and CONVENTIONS.md (j). This shard starts the Gaussian-boson block.

24.2 Problem Statement

We now restrict the lattice-symmetry problem to a solvable class: translation-invariant Gaussian bosonic lattice systems in spatial dimensions

$$d = 1, 2, 3, \qquad D = d + 1.$$

The physical target is Lorentz/Poincare covariance in the scaling limit. The expected endpoint is the scalar free field:

$$\omega(k)^2 = m^2 + |k|^2$$

after the correct normalization of lattice spacing, velocity, field strength, and mass.

The inputs are not merely a matrix Hamiltonian. The compiler receives:

$\Lambda_{\epsilon,L}$	periodic hypercubic lattice or infinite lattice,
q_x, p_x	canonical bosonic field variables,
V_r	translation-invariant quasi-local quadratic coefficients,
ϵ	lattice spacing,
S_ϵ	state/scaling-map data, when a continuum claim is requested.

The first coefficient tier is the scalar canonical Hamiltonian

$$H = \frac{1}{2} \sum_x p_x^2 + \frac{1}{2} \sum_{x,r} q_x V_r q_{x+r}.$$

Full bosonic BdG/pairing systems are a later tier, not silently included in the first convention.

24.3 North Star

For each admissible coefficient family $V_r^{(\epsilon)}$, compute the candidate generators as functions of the coefficients:

$$H, \quad P_a, \quad J_{ab}, \quad K_a,$$

then compute the residuals in the Poincare algebra. The residuals become conditions on the symbol

$$\omega_\epsilon(k)^2 = \sum_r V_r^{(\epsilon)} e^{i\epsilon k \cdot r}.$$

The end goal is a theorem-shaped statement:

$$\begin{gathered}\text{Gaussian lattice coefficients satisfying locality, positivity,}\\ \text{single low-energy minimum, and isotropic Klein-Gordon Taylor data}\\ \Longrightarrow \text{ scaling limit is the free scalar/Klein-Gordon Poincare system.}\end{gathered}$$

CA-15 proof boundary. This is a theorem-shaped target, not a theorem already obtained from the symbol checks. A continuum claim still needs the CA-15 payload: scaling maps, state sequence, topology, operator domains, local observable covariance, and boundary/renormalization policy. The value of the Gaussian block is that the algebraic residuals can be made explicit before attempting that proof.

24.4 Shard Roadmap

The current Gaussian block is:

CA-24. Symbol calculus for the scalar coefficient tier, including the boost-time residual and the checked nearest-neighbour Klein-Gordon seed.

CA-25. Diagonalization into one-particle symbols, with the sourced free-scalar Fock construction separated from the local coefficient-symbol derivation and BdG pairing deferred.

CA-26. Smooth-patch one-particle generator algebra for H, P, J, K, exact brackets, dispersion-dependent residuals, and only formal $\mathrm{d}\Gamma$ bookkeeping on the algebraic finite-particle core.

CA-27. Lorentz residual conditions on coefficients: Taylor constraints, anisotropic failures, and doubler counterexamples.

CA-28. Numerical invariant suite for scalar symbols and finite periodic stiffness matrices, covering Klein-Gordon, anisotropic, and doubler examples.

CA-29. Real-space energy-density convention for the scalar Gaussian Hamiltonian: cell volume, bond sharing, boundary policy, vacuum subtraction, and improvement terms before any position-weighted use.

The queued follow-up block is proposal work, not completed stress-energy material:

1+1 stress candidates. Build candidate lattice stress-energy formulas only after the CA-29 density and a registered current convention are in place.

Current-vs-symbol equivalence. Check when real-space first moments and currents reproduce the flat local-symbol residuals used in CA-24–CA-27.

Higher-dimensional cells. Propose cell-current formulas in $d = 2, 3$, keeping orientation, boundary, and improvement choices explicit.

Stress-energy numerical suite. Add numerical checks only after candidate formulas and source-backed expected identities exist.

Source acquisition. Register the continuum stress-energy/Poincare-generator source, harmonic-chain energy-current formulas, and the discrete Gaussian/free-field Virasoro source.

24.5 Side Quests

Source acquisition. The registered local OAR sources cover the free scalar lattice scaling limit and spacetime translations. They explicitly warn that Lorentz/conformal convergence requires further work. We need a registered source for the exact continuum stress-energy/Poincare-generator formulas, and a registered copy of the discrete Gaussian free-field Virasoro source currently present only as bibliography metadata.

Generator ambiguity. A local Hamiltonian density is not unique:

$$h_x \mapsto h_x + \nabla \cdot b_x$$

changes first moments by boundary or current terms. The Gaussian block must track which density convention produced $K_a = \sum_x x_a h_x$.

Periodic positions. Numerical tests use periodic lattices and finite Brillouin grids. First moments, position coordinates, and finite-grid momentum-coordinate generators need a branch, sawtooth, Fourier interpolation, or another explicit periodic coordinate convention. Until that is fixed, early finite tests are symbol/eigenvalue checks on the periodic grid, or open-chain first-moment checks.

BdG pairing. A general quasi-local Hamiltonian in creation and annihilation operators may include pairing terms. The scalar q, p tier captures Klein-Gordon directly; BdG requires symplectic positivity and a stable bosonic Bogoliubov convention before generator formulas are promoted.

24.6 Numerical Verification Plan

The Julia layer should stay at the invariant level:

1. symbol from coefficients equals the analytic dispersion,
2. $\frac{1}{2}\nabla\omega_\epsilon(k)^2$ equals the smooth-patch boost-time residual,
3. $\frac{\sin(\epsilon k_a)}{\epsilon} \to k_a$ for the KG lattice,
4. anisotropic Hessians fail the Lorentz residual checks,
5. finite Fourier matrices diagonalize periodic stiffness matrices.

The first three checks are implemented with CA-24.

24.7 Acceptance Boundary

The roadmap rows above remain scoped by the CA-15 proof boundary stated next to the theorem-shaped goal. Passing symbol or finite-matrix tests is not yet a proof of Lorentz invariance.

25 Gaussian Boson Symbol Calculus and the Boost-Time Residual

25.1 Status, Sources, and Dependencies

Status: sourced model class plus checked flat local derivation. The standard lattice scalar-field Hamiltonian and dispersion are sourced. The boost-time residual calculation is a local one-particle derivation on flat $L^2(dk)$, checked by Julia on positive-dispersion patches. It is not yet a sourced OAR or mass-shell Poincare representation theorem, and massless zero-mode examples are coefficient-level tests only.

Dependencies: CA-11, CA-12, CA-23, and CONVENTIONS.md (h), (j), (k).

25.2 Coefficient Tier

Fix $d \in \{1, 2, 3\}$. On a periodic hypercubic lattice with spacing ϵ, use canonical fields

$$[q_x, q_y] = 0, \qquad [p_x, p_y] = 0, \qquad [q_x, p_y] = i\delta_{xy}.$$

The first scalar Gaussian tier is

$$H = \frac{1}{2}\sum_x p_x^2 + \frac{1}{2}\sum_{x,r} q_x V_r q_{x+r}, \qquad V_r = V_{-r} \in \mathbb{R}.$$

Quasi-locality means that V_r has enough decay, or finite range in the first examples, so that the Fourier symbol and its derivatives used below exist.

Define

$$\omega_\epsilon(k)^2 := \sum_r V_r e^{i\epsilon k\cdot r}.$$

For a stable scalar model, this symbol is non-negative, with the zero-mode policy stated separately in the massless case. The helper parameter m is a nonnegative mass label; the generator statements below use only the stricter case $\omega_\epsilon(k) > 0$ on the chosen patch.

25.3 Nearest-Neighbour Klein-Gordon Example

The sourced lattice scalar-field example has, in the notation of this shard,

$$V_0 = m^2 + \frac{2d}{\epsilon^2}, \qquad V_{\pm e_a} = -\frac{1}{\epsilon^2}.$$

Therefore

$$\begin{aligned}\omega_{\epsilon,m}(k)^2 &= m^2 + \frac{2d}{\epsilon^2} - \frac{1}{\epsilon^2}\sum_{a=1}^{d}(e^{i\epsilon k_a} + e^{-i\epsilon k_a}) \\ &= m^2 + \frac{2}{\epsilon^2}\sum_{a=1}^{d}(1 - \cos(\epsilon k_a)).\end{aligned}$$

For fixed physical k and $\epsilon \to 0$,

$$\omega_{\epsilon,m}(k)^2 = m^2 + |k|^2 + O(\epsilon^2|k|^4).$$

25.4 Momentum-Space Position

After diagonalization, this shard uses the flat local momentum space $L^2(\mathcal{U}, dk)$. The one-particle Hamiltonian is multiplication by $\omega(k)$, with $\omega(k) > 0$ on $\mathcal{U}$, and

$$X_a = i\partial_{k_a}$$

on a common smooth core. The time-zero boost candidate corresponding to the energy first moment is the flat-measure symmetrized operator

$$K_a = \frac{1}{2}(X_a\omega + \omega X_a).$$

The symmetrization is the one-particle version of taking a self-adjoint first moment of the energy density. If the one-particle scalar product is changed, the adjoint and hence the boost formula must be rederived; no unitary equivalence with the OAR weighted space or Schottenloher mass-shell realization is asserted here.

25.5 Boost-Time Residual

For a test wavefunction ψ,

$$\begin{aligned}[K_a, \omega]\psi &= \frac{1}{2}(X_a\omega^2 - \omega X_a\omega)\psi \\ &= i\,\omega\,(\partial_a\omega)\psi.\end{aligned}$$

Thus

$$i[\omega, K_a] = \omega\,\partial_a\omega = \frac{1}{2}\partial_a\omega(k)^2.$$

This is the algebraic output of the boost-time commutator. With the vector-field-to-Stone sign map of CONVENTIONS.md (k), the continuum Poincare target is $i[H, K_a] = P_a$. Taking $P_a = k_a$, one needs

$$\frac{1}{2}\partial_a\omega(k)^2 = k_a.$$

Integrating over k gives

$$\omega(k)^2 = m^2 + |k|^2$$

on the connected low-energy patch. With speed c, the right side is c^2k_a, giving $\omega(k)^2 = m^2 + c^2|k|^2$.

25.6 Lattice Residual

For the nearest-neighbour lattice Klein-Gordon symbol,

$$\frac{1}{2}\partial_a\omega_{\epsilon,m}(k)^2 = \frac{\sin(\epsilon k_a)}{\epsilon}.$$

Therefore the boost-time commutator has the correct continuum limit:

$$\frac{\sin(\epsilon k_a)}{\epsilon} = k_a + O(\epsilon^2 k_a^3).$$

The Julia tests check this in $d = 1, 2, 3$, both from the closed formula and by evaluating the coefficient symbol $\sum_r V_r e^{i\epsilon k\cdot r}$.

25.7 Rotation Residual

For the one-particle rotation candidate

$$J_{ab} = X_a P_b - X_b P_a, \qquad P_a = k_a,$$

one obtains

$$i[\omega, J_{ab}] = k_b \partial_a \omega - k_a \partial_b \omega.$$

Thus rotations commute with the Hamiltonian exactly when the dispersion is radial on the momentum region under consideration. For a hypercubic lattice this is not exact over the full Brillouin zone, but the Klein-Gordon expansion is isotropic to quadratic order at $k = 0$.

25.8 Boundary

This shard is a one-particle symbol calculation. The many-body generators are second quantizations plus vacuum-energy and domain policies. Real-space locality of $K_a = \sum_x x_a h_x$, boundary terms, and convergence of the second-quantized operators are deferred to later shards.

26 Gaussian Boson Diagonalization and the One-Particle Symbol

26.1 Status, Sources, and Dependencies

Status: sourced template plus flat local coefficient derivation. The standard lattice free-scalar Hamiltonian, Fock representation, continuum one-particle space, and second-quantized Hamiltonian are sourced. The extension from nearest-neighbour coefficients to an arbitrary real translation-invariant scalar symbol is a local algebraic calculation on the flat $L^2(dk)$ symbol space fixed in CONVENTIONS.md (j).

Dependencies: CA-23–CA-24 and CONVENTIONS.md (a), (f), (j).

26.2 Scalar Coefficient Tier

Fix $d \in \{1, 2, 3\}$. The first Gaussian tier is

$$H = \frac{1}{2}\sum_x p_x^2 + \frac{1}{2}\sum_{x,r} q_x V_r q_{x+r}, \qquad V_r = V_{-r} \in \mathbb{R}.$$

On an infinite translation-invariant lattice, or on a periodic lattice before taking volume limits, the Fourier transform sends the potential kernel to

$$\omega_\epsilon(k)^2 = \sum_r V_r e^{i\epsilon k\cdot r}.$$

The reality condition $V_r = V_{-r}$ makes this symbol real:

$$\omega_\epsilon(k)^2 = V_0 + \sum_{r>0} 2V_r \cos(\epsilon k \cdot r).$$

The stability condition for this tier is

$$\omega_\epsilon(k)^2 \geq 0$$

on the Brillouin zone. The `mass` parameter is a nonnegative label; in the nearest-neighbour Klein-Gordon family, $m > 0$ gives the positive-dispersion case used by the current generator statements. For a general coefficient symbol the required condition remains $\omega_\epsilon(k) > 0$ on the chosen patch. When $m = 0$, zero modes are coefficient data only until a zero-mode convention excludes or otherwise controls them.

26.3 Mode Decoupling

The Hamilton equations are a local derivation from the canonical commutators:

$$\dot{q}_x = p_x, \qquad \dot{p}_x = -\sum_r V_r q_{x+r}.$$

Fourier transforming gives

$$\dot{\widehat{q}}(k) = \widehat{p}(k), \qquad \dot{\widehat{p}}(k) = -\omega_\epsilon(k)^2 \widehat{q}(k),$$

and hence

$$\ddot{\widehat{q}}(k) + \omega_\epsilon(k)^2 \widehat{q}(k) = 0.$$

Thus each momentum mode is an oscillator with frequency $\omega_\epsilon(k)$. This is the algebraic reason the compiler can reduce the scalar Gaussian tier to multiplication by a dispersion symbol.

26.4 Complex Structure and Fock Space

For $\omega > 0$ on the chosen momentum patch, the oscillator complex structure is

$$(q, p) \longmapsto (-\omega p, \omega^{-1} q).$$

The sourced free-scalar construction uses this pattern with $\gamma_m(k) = \sqrt{|k|^2 + m^2}$ in the continuum and with $\gamma_m^{(N)}(k)$ on the lattice. It also realizes the Weyl algebra in a bosonic Fock representation by creation and annihilation operators. The source scalar product is the OAR one on real Cauchy data, with $\gamma_m^{-1/2}\widehat{q} + i\gamma_m^{1/2}\widehat{p}$; the text then realizes the representation on a mass-independent flat L^2 Fock space. This shard uses only the resulting flat symbol bookkeeping and does not prove that the boost formula of CA-24/CA-26 intertwines the OAR or mass-shell representation.

The compiler-level output for the positive-dispersion scalar tier is therefore:

$\mathfrak{h}_\epsilon$	flat one-particle momentum space on the chosen patch or grid,
h_ϵ	multiplication by $\omega_\epsilon(k)$,
$\mathcal{F}_+(\mathfrak{h}_\epsilon)$	symmetric Fock completion,
H_ϵ	$d\Gamma(h_\epsilon) + E_{\epsilon,0}$ if a vacuum-energy policy is named.

The additive constant $E_{\epsilon,0}$ is not part of the commutator algebra for the generators considered here, but it is part of a Hamiltonian-density normal-ordering policy and must be recorded before using ground-state energies as numerical claims.

26.5 What Is Not Yet Included

This shard does not define the full bosonic BdG tier. A general quadratic Hamiltonian in creation and annihilation operators may include pairing terms, and then the correct diagonalization is a symplectic/Bogoliubov problem with positivity, implementability, and domain conditions. Those conventions are not fixed here.

The current compiler rule is deliberately narrower:

$$\text{scalar } (q, p) \text{ tier} \quad \Rightarrow \quad \omega_\epsilon(k)^2 \text{ symbol} \quad \Rightarrow \quad h_\epsilon = \omega_\epsilon(k).$$

BdG inputs must fail closed or route to a later convention until that tier is written.

26.6 Compiler Consequence

For the positive-dispersion scalar Gaussian block, later candidate generators can be constructed first on the one-particle smooth momentum core and then second-quantized. The coefficient data enter only through $\omega_\epsilon(k)^2$ and its derivatives, while the continuum proof still requires the CA-15 payload: scaling maps, state convergence, domains, and observable covariance.

27 One-Particle Generator Algebra for Gaussian Bosons

27.1 Status, Sources, and Dependencies

Status: flat local derivation with limited Julia checks. The Poincare target, Stone convention, and free-boson Fock realization are sourced. The commutator identities in this shard are local differential-operator derivations on a flat $L^2(dk)$ smooth momentum-space core. The Julia suite currently checks coefficient/symbol consequences: Klein-Gordon symbols, boost-time symbols, sampled rotation-Hamiltonian residuals, sampled boost-boost residual coefficient data, low-energy Hessian examples, finite periodic massive spectra, the massless doubler coefficient-rejection witness, and the Gaussian tolerance policy. It does not check the full generator bracket table, finite coordinate commutators, or any second-quantized identity. The smooth-core derivation is not yet a sourced OAR or mass-shell Poincare representation theorem.

Dependencies: CA-11, CA-24–CA-25, and CONVENTIONS.md (h), (j), (k).

27.2 Momentum Core

There are two separate settings.

Smooth patch. Work on a connected smooth momentum patch $\mathcal{U}$ where $\omega(k) > 0$, with the flat measure dk. Let

$$\mathcal{D}_0 = C_c^\infty(\mathcal{U}) \subset L^2(\mathcal{U}, dk)$$

and define

$$X_a = i\partial_{k_a}, \qquad h = \omega(k), \qquad p_a = k_a.$$

The one-particle Hamiltonian and momenta are multiplication operators:

$$H^{(1)} = h, \qquad P_a^{(1)} = p_a.$$

The rotation and time-zero boost candidates are

$$J_{ab}^{(1)} = X_a P_b - X_b P_a, \qquad K_a^{(1)} = \frac{1}{2}(X_a h + h X_a).$$

These are written in the self-adjoint sign convention of CONVENTIONS.md (k); in particular J_{ab} is the Gaussian rotation sign, not the raw lowered-index CA-11 vector field M_{ab}. All commutators below are identities on $\mathcal{D}_0$; promoting them to self-adjoint generator statements is a separate domain theorem. Promoting this flat formula to the OAR one-particle Hilbert space or Schottenloher's mass-shell/free-boson Poincare representation additionally requires a sourced or locally derived unitary intertwiner.

Finite periodic grids. On a finite periodic Gaussian lattice, the dual grid is used for multiplication symbols and Fourier eigenvalue checks. The current finite suite does not place the differential operator $X_a = i\partial_{k_a}$ on that grid and does not claim an exact finite $[X_a, p_b] = i\delta_{ab}$ commutator. Periodic first-moment or momentum-coordinate generator tests require an explicit branch, sawtooth, Fourier-interpolation, or comparable periodic-coordinate convention before they can be stated. Thus the exact differential commutators below are smooth-patch algebra; the finite periodic checks remain symbol/eigenvalue checks.

27.3 Exact Translation and Rotation Brackets

On the smooth core, because h and p_a are multiplication operators,

$$i[h, p_a] = 0.$$

The canonical differential identity is the smooth-patch identity

$$[X_a, p_b] = i\delta_{ab}.$$

Therefore

$$i[P_c, J_{ab}] = \delta_{ac}P_b - \delta_{bc}P_a.$$

This is independent of the dispersion. It is kinematical on the smooth patch: it comes from the chosen momentum coordinate and the differential representation of rotations, in the rotation sign convention of CONVENTIONS.md (k).

27.4 Boost-Momentum Bracket

For the boost candidate,

$$\begin{aligned}[P_b, K_a] &= \frac{1}{2}\big([P_b, X_a]h + h[P_b, X_a]\big) \\ &= -i\delta_{ab}h.\end{aligned}$$

Hence

$$i[P_b, K_a] = \delta_{ab}H.$$

This is the boost-momentum line of the self-adjoint target table in CONVENTIONS.md (k). On the smooth patch, this part of the Poincare algebra is exact for every positive scalar dispersion because it only uses K_a as the symmetrized first moment of energy.

27.5 Boost-Time Residual

CA-24 derived

$$i[H, K_a] = \frac{1}{2}\partial_a\omega(k)^2.$$

The continuum target with speed $c = 1$ wants $P_a = k_a$. The residual is

$$R_a(k) := \frac{1}{2}\partial_a\omega(k)^2 - k_a.$$

For speed c, replace k_a by c^2k_a. This is the first algebraic filter on Hamiltonian coefficients. Julia now exposes it both as the boost-time symbol $B_a = \frac{1}{2}\partial_a\omega^2$ and as the residual vector $R_a = B_a - c^2k_a$, still only at coefficient-symbol level.

27.6 Rotation-Hamiltonian Residual

The Hamiltonian commutes with rotations iff the dispersion is radial on the patch:

$$i[h, J_{ab}] = k_b\partial_a\omega - k_a\partial_b\omega.$$

Equivalently,

$$k_b\partial_a\omega^2 - k_a\partial_b\omega^2 = 0$$

where $\omega > 0$. Hypercubic lattice symbols usually fail this globally on the Brillouin zone but may satisfy it to leading order at a low-energy minimum. The Julia helper records the sampled antisymmetric matrix

$$2(k_bB_a - k_aB_b), \qquad B_a = \frac{1}{2}\partial_a\omega^2,$$

so the factor of two follows the boost-time convention instead of being re-derived independently.

27.7 Boost-Boost Residual

Write

$$B_a(k) := \frac{1}{2}\partial_a\omega(k)^2, \qquad R_a(k) := B_a(k) - k_a.$$

Equivalently $B_a = h\partial_a h$. Since

$$K_a = i\left(h\partial_a + \frac{1}{2}\partial_a h\right),$$

the commutator of the first-order differential parts gives

$$i[K_a, K_b] = J_{ab} + R_b X_a - R_a X_b.$$

The leading $+J_{ab}$ is the sign dictated by CONVENTIONS.md (k): CA-11 has $[M_{0a}, M_{0b}] = -M_{ab}$, while the self-adjoint rotation here is $J_{ab} = -A_{M_{ab}}$. Here the possible divergence term vanishes because R is a gradient residual: $\partial_a R_b = \partial_b R_a$. Thus the boost-boost bracket closes to the rotation bracket whenever the boost-time residual R_a vanishes on the patch. The Julia helper records only the sampled coefficient tensor for the first-order residual $R_b X_a - R_a X_b$: its (a, b, c) entry is the coefficient of X_c for the ordered boost pair (a, b). It is not a finite-grid coordinate commutator and does not check the boost-boost operator bracket. The remaining caveat is analytic, not algebraic: all of these are unbounded operators and still require a common self-adjointness/core theorem before this becomes a continuum representation statement.

27.8 Second Quantization

Let

$$\mathcal{F}_{\mathrm{alg}}(\mathcal{D}_0) := \bigoplus_{n=0}^{\mathrm{fin}} \mathcal{D}_0^{\otimes_s n} \subset \mathcal{F}_+\big(L^2(\mathcal{U}, dk)\big)$$

be the algebraic finite-particle core. The formal many-body candidates are

$$H = d\Gamma(h), \quad P_a = d\Gamma(p_a), \quad J_{ab} = d\Gamma(J_{ab}^{(1)}), \quad K_a = d\Gamma(K_a^{(1)}),$$

with a separate vacuum-energy convention if absolute Hamiltonian values are used. The formal core identity is

$$[d\Gamma(A), d\Gamma(B)]\psi = d\Gamma([A, B])\psi, \qquad \psi \in \mathcal{F}_{\mathrm{alg}}(\mathcal{D}_0),$$

whenever the one-particle commutator identity is already established on $\mathcal{D}_0$ and the displayed products preserve that algebraic domain. This is a local $d\Gamma$ bookkeeping statement on the algebraic finite-particle core, not a Julia-checked invariant and not a self-adjoint many-body representation theorem. The current tests do not lift residuals to Gaussian Fock space; domain/self-adjointness for the unbounded $d\Gamma$ generators and the sourced OAR/mass-shell bridge remain open proof obligations.

28 Lorentz Residual Conditions on Gaussian Coefficients

28.1 Status, Sources, and Dependencies

Status: checked local coefficient calculus. The sourced massive free scalar model supplies the target dispersion. The general coefficient residuals are local Taylor calculations from $\omega_\epsilon(k)^2 = \sum_r V_r e^{i\epsilon k\cdot r}$. The examples in this shard are checked in Julia; the massless doubler is only a coefficient-level rejection witness, not a positive-dispersion generator example.

Dependencies: CA-24–CA-26 and CONVENTIONS.md (j).

28.2 Gradient Residual

For a scalar coefficient family,

$$\omega_\epsilon(k)^2 = \sum_r V_r e^{i\epsilon k\cdot r}.$$

The boost-time residual from CA-26 is

$$R_a(k) = \frac{1}{2}\partial_a \omega_\epsilon(k)^2 - c^2 k_a.$$

Explicitly,

$$\frac{1}{2}\partial_a \omega_\epsilon(k)^2 = \frac{i\epsilon}{2}\sum_r r_a V_r e^{i\epsilon k\cdot r}.$$

When $V_r = V_{-r}$, this is real and can be written as a sine sum:

$$\frac{1}{2}\partial_a \omega_\epsilon(k)^2 = -\epsilon \sum_{r>0} r_a V_r \sin(\epsilon k\cdot r),$$

where $r > 0$ denotes any fixed choice of one representative from each $\{r, -r\}$ pair.

28.3 Low-Energy Hessian Condition

At a symmetric minimum $k = 0$, the first derivative vanishes. The Hessian is

$$\partial_a\partial_b \omega_\epsilon(0)^2 = -\epsilon^2 \sum_r r_a r_b V_r.$$

Therefore the quadratic Lorentz-speed condition is

$$-\frac{\epsilon^2}{2}\sum_r r_a r_b V_r = c^2 \delta_{ab}.$$

This is the first finite-dimensional coefficient test. In compiler form:

$$\mathsf{HessResidual}_{ab} := -\frac{\epsilon^2}{2}\sum_r r_a r_b V_r - c^2\delta_{ab}.$$

It must vanish at the chosen low-energy point before the model can be routed to a Klein-Gordon continuum target.

It is not a positivity or patch certificate. The Hessian only sees the Taylor coefficient at the chosen point; a finite-grid symbol can still be negative elsewhere or have additional low-energy minima.

28.4 Nearest-Neighbour Klein-Gordon

For

$$V_0 = m^2 + \frac{2d}{\epsilon^2}, \qquad V_{\pm e_a} = -\frac{1}{\epsilon^2},$$

one obtains

$$-\frac{\epsilon^2}{2}\sum_r r_a r_b V_r = \delta_{ab}.$$

The gradient residual is the stronger finite-ϵ expression

$$R_a(k) = \frac{\sin(\epsilon k_a)}{\epsilon} - k_a,$$

which vanishes as $O(\epsilon^2 k_a^3)$ for fixed physical momentum.

28.5 Anisotropic Counterexample

Let the nearest-neighbour coefficients carry different speeds c_a:

$$\omega_\epsilon(k)^2 = m^2 + \frac{2}{\epsilon^2}\sum_a c_a^2(1 - \cos(\epsilon k_a)).$$

Then

$$-\frac{\epsilon^2}{2}\sum_r r_a r_b V_r = c_a^2 \delta_{ab}.$$

Unless all c_a agree with the target speed, the boost-time residual cannot match a single Lorentz cone. The Julia test suite checks this as a deliberate failure case, not as an approximate success.

28.6 Hessian-Passing Instability Witness

In $d = 1$, at $\epsilon = 1$, consider the scalar coefficients

$$V_0 = 0, \qquad V_{\pm 1} = 1, \qquad V_{\pm 2} = -\frac{1}{2}.$$

The symbol is

$$\omega(k)^2 = 2\cos k - \cos(2k).$$

Its low-energy Hessian data pass the speed-one condition:

$$-\frac{1}{2}\sum_r r^2 V_r = 1.$$

However, the $L = 4$ periodic sample has values

$$\omega(0)^2 = 1, \quad \omega(\pi/2)^2 = 1, \quad \omega(\pi)^2 = -3, \quad \omega(3\pi/2)^2 = 1.$$

The checked minimum-data helper records the negative minimum -3, with uncentered label 2 and momentum π, plus centered label -2 and momentum $-\pi$. Enabling its nonnegative finite-grid gate makes the helper raise an error. Thus the Hessian residual can vanish while the finite periodic symbol is unstable.

28.7 Doubler Counterexample

The Hessian condition is necessary but not sufficient. The symbol

$$\omega_\epsilon(k)^2 = m^2 + \frac{1}{\epsilon^2}\sum_a \sin^2(\epsilon k_a)$$

has the correct quadratic Hessian at $k = 0$ when $c = 1$. The checked example sets $m = 0$, so it is massless and has zero modes; by CONVENTIONS.md (j), this is outside the current positive-dispersion generator theorem. Its role is coefficient-level rejection: on even periodic grids it has multiple minima at Brillouin-zone points with $\sin(\epsilon k_a) = 0$. Therefore the compiler also needs a *single-patch* or *sector-selection* condition before a continuum relativistic scalar target is attached.

The current tests record this in $d = 1, 2, 3$ on the $L_a = 4$ periodic grids: the Hessian residual still vanishes at speed 1, while `symbol_minimum_data` finds 2^d zero minima with each centered label coordinate in $\{0, -2\}$. This is a finite-grid witness of extra patches, not a continuum species-count theorem.

28.8 Compiler Rule

For the scalar Gaussian tier, a Lorentz-target compile attempt must emit at least:

mass term	$\omega(0)^2$,
speed matrix	$-\frac{\epsilon^2}{2}\sum_r r_a r_b V_r$,
boost residual	$R_a(k)$,
rotation residual	$k_b\partial_a\omega^2 - k_a\partial_b\omega^2$,
boost-boost residual	$R_b X_a - R_a X_b$,
patch data	finite-grid minimum value, labels, momenta, and count.

Only the mass and speed entries are finite Taylor data. The rotation and boost-boost entries are now sampled smooth-symbol residuals in Julia, with the boost-boost entry represented as first-order X_c-coefficient data rather than as a finite-grid operator commutator. The residual and patch entries are what prevent an anisotropic or doubler model from being mislabeled as a single Klein-Gordon field.

29 Numerical Verification Suite for Gaussian Boson Examples

29.1 Status, Sources, and Dependencies

Status: checked Julia shard. This shard records the numerical invariants now implemented for the scalar Gaussian tier. The checks are not continuum proofs; they are finite algebraic witnesses that the coefficient compiler is computing the intended symbols and residuals. The massless doubler check is a coefficient-level rejection test, not a Fock-generator check. The finite periodic checks below do not assert an exact differential $[X_a, P_b] = i\delta_{ab}$ commutator.

The current numerical surface is deliberately narrow: it checks scalar coefficient structure, finite-periodic symbol values, finite stiffness spectra, assigned Fourier eigenvectors, periodic minimum data, and the implemented coefficient residuals/Hessians, including sampled rotation-Hamiltonian and boost-boost residual data on smooth-symbol momenta. It also checks centered periodic momentum labels for branch-aware low-energy windows, a loud finite-grid nonnegativity gate when requested, and test-local central finite-difference boost-time derivative oracles. It does not test finite boost generators, finite rotation generators, coordinate-generator commutators, or full generator closure.

Dependencies: CA-24–CA-27, INDEX.md, and CONVENTIONS.md (j).

29.2 Why These Tests

The numerical layer is not allowed to assert that a plot "looks relativistic". Each check must compare two independently constructed objects:

$$\text{real-space coefficient data} \quad \longleftrightarrow \quad \text{momentum-space symbol data.}$$

For Gaussian systems this is unusually powerful because the finite periodic stiffness matrix is exactly diagonalized by Fourier modes.

The boost-time derivative oracle follows the same rule. The production boost-time helper is compared to $\frac{1}{2}\nabla_k \omega(k)^2$ obtained by central finite differences of the separate scalar-symbol evaluator. Those tests use generic symmetric finite-range coefficient sets in $d = 1, 2, 3$, with off-axis terms in the higher-dimensional cases and sample momenta away from high-symmetry points.

The rotation and boost-boost residual tests are also smooth-symbol checks. They sample coefficient formulas at named momenta and compare residual sizes or tensor entries; they do not introduce $X_a = i\partial_{k_a}$ as a finite periodic matrix.

29.3 Numerical Tolerance Policy

CONVENTIONS.md (j) fixes the tolerance names used by this shard. With the `GAUSSIAN_` prefix and `_ATOL/_RTOL` suffix understood, the default pairs are

`SYMBOL_IMAG`	$(10^{-10}, 10^{-10})$,
`SYMBOL_VALUE`	$(10^{-10}, 10^{-10})$,
`EIGENVALUE`	$(10^{-10}, 10^{-10})$,
`MINIMUM_COUNT`	$(10^{-10}, 10^{-10})$,
`SMALL_SPACING_RESIDUAL`	$(5 \cdot 10^{-4}, 10^{-10})$.

The symbol-imaginary validator compares the complex symbol with its real part after the dimension and spacing preconditions are checked. The small-spacing pair is a named finite-sample tolerance, not a continuum error bound. The finite-difference derivative oracle is test-local and uses $(2\cdot 10^{-8}, 2\cdot 10^{-8})$; it is not part of the production tolerance surface.

29.4 Structural Coefficient Validation

The scalar Gaussian helpers now fail closed by default on coefficient data outside CONVENTIONS.md (j). The validator requires nonempty data, integer offsets, a common dimension $d \in \{1, 2, 3\}$, real coefficients, and aggregate paired equality $V_r = V_{-r}$ after duplicate offsets are summed. Negative tests cover a missing partner, unequal paired aggregates, complex coefficients, dimension mismatch, and non-integer offsets.

29.5 Periodic Stiffness Matrix

For finite periodic sizes $L_1, \ldots, L_d$, define the stiffness matrix

$$V_{xy} = \sum_r V_r \, \mathbf{1}_{y=x+r \bmod L}.$$

For the Fourier vector

$$\varphi_k(x) = |\Lambda|^{-1/2} e^{i\epsilon k \cdot x},$$

one has the exact identity

$$(V\varphi_k)(x) = \left(\sum_r V_r e^{i\epsilon k \cdot r} \right) \varphi_k(x) = \omega_\epsilon(k)^2 \varphi_k(x).$$

Therefore each dual label n gives a tested eigenvector with its assigned symbol value, and the sorted eigenvalues of the finite stiffness matrix must equal the sorted symbol values on the periodic Brillouin grid. These finite Fourier invariants are checked in Julia for $d = 1, 2, 3$, including mixed odd/even sizes with every $L_a > 2R$ and an off-axis symmetric pair $(1, 1), (-1, -1)$. They are not finite periodic first-moment or momentum-coordinate commutator tests; those require a branch, sawtooth, Fourier-interpolation, or other coordinate convention before they can become numerical claims.

The eigenvector suite also includes a one-sided shift sentinel to pin the implementation direction $y = x + r$. That sentinel is not a scalar Gaussian Hamiltonian claim. The strict default rejects it, and the test reaches this low-level matrix path only through an explicit non-validating keyword.

The eigenvalue oracle deliberately keeps the uncentered grid helper, because the tested symbol spectra are branch-insensitive. The separate centered grid helper returns the centered integer dual label and its physical representative

$$k_a = \frac{2\pi n_a}{\epsilon L_a}.$$

For even $L = 2r$, the branch is $n_a \in \{-r, \ldots, 0, \ldots, r - 1\}$, matching the centered dual lattice in the OAR source. Thus, at $L = 4$ and $\epsilon = 1$, the uncentered representative $3\pi/2$ is recorded on the centered branch as $n = -1$ and $k = -\pi/2$. This is only a momentum-label helper; it is not a periodic position-coordinate or generator construction.

29.6 Implemented Example Systems

Nearest-neighbour Klein-Gordon. The tests check the coefficient symbol against

$$m^2 + \frac{2}{\epsilon^2} \sum_a (1 - \cos(\epsilon k_a)),$$

the boost-time symbol against $\sin(\epsilon k_a)/\epsilon$, the residual vector against $\sin(\epsilon k_a)/\epsilon - k_a$, and the low-energy Hessian residual against zero. The sampled rotation and boost-boost residual helpers are small on fixed low-energy momenta as ϵ is made small. The boost-time helper is also checked on non-Klein-Gordon finite-range symbols by the independent finite-difference oracle above.

Anisotropic cones. The tests build coefficients with speeds (c_1, c_2, c_3). They verify that the Hessian is

$$2\,\mathrm{diag}(c_1^2, c_2^2, c_3^2)$$

and that the Lorentz residual is nonzero when the target speed is 1. A separate sampled residual test checks that anisotropy also gives nonzero rotation-Hamiltonian and boost-boost residual data at a generic smooth momentum.

Hessian-passing unstable scalar. The tests build the one-dimensional coefficient witness

$$V_0 = 0, \qquad V_{\pm 1} = 1, \qquad V_{\pm 2} = -\frac{1}{2}.$$

Its speed-one Hessian residual vanishes, but finite periodic sampling at $L = 4$ has minimum -3. The minimum-data helper records one minimum at location 3, uncentered label 2, centered label -2, uncentered momentum π, and centered momentum $-\pi$. Requesting a nonnegative finite-grid symbol raises an error, so this example cannot pass as a stable scalar symbol.

Doubler symbol. The tests build

$$\omega_\epsilon(k)^2 = m^2 + \epsilon^{-2} \sum_a \sin^2(\epsilon k_a)$$

with $m = 0$. In $d = 1, 2, 3$, the speed-one Hessian residual vanishes on the chosen patch, but the $L_a = 4$ periodic grid has 2^d zero minima with centered label coordinates in $\{0, -2\}$. This catches the case where the Hessian at one minimum is correct but the compiler would need a multi-species, sector-selection, or zero-mode policy output. It is deliberately outside the positive-dispersion generator statements. The J6 symbol-residual test also samples a doubler momentum where the rotation residual is globally nonzero and an extra zero where R_a is nonzero relative to the selected $k = 0$ branch.

29.7 Current Julia Surface

The checked helper layer is:

`kg_nearest_neighbor_coefficients`	standard scalar KG coefficients,
`anisotropic_kg_nearest_neighbor_coefficients`	anisotropic coefficient witness,
`doubler_quadratic_coefficients`	coefficient-level doubler witness,
`validate_scalar_coefficients`	structural scalar-coefficient validator,
`scalar_quadratic_symbol`	$\sum_r V_r e^{ik\cdot r}$,
`boost_time_symbol_from_coefficients`	$\frac{1}{2}\nabla\omega^2$,
`boost_time_residual_from_coefficients`	$\frac{1}{2}\nabla\omega^2 - c^2 k$,
`rotation_hamiltonian_residual_from_coefficients`	$k_b\partial_a\omega^2 - k_a\partial_b\omega^2$,
`boost_boost_residual_coefficients_from_coefficients`	coefficients of $R_b X_a - R_a X_b$,
`low_energy_hessian_from_coefficients`	$\partial_a\partial_b\omega^2(0)$,
`lorentz_hessian_residual`	$\frac{1}{2}\mathrm{Hess}(\omega^2)(0) - c^2 I$,
`periodic_stiffness_matrix`	finite periodic real-space matrix,
`periodic_momentum_grid`	uncentered spectrum grid,
`centered_periodic_momentum_grid`	centered labels and momenta,
`periodic_fourier_vector`	Fourier vector for a dual label,
`periodic_symbol_values`	finite Brillouin-grid eigenvalue oracle,
`symbol_minimum_data`	finite-grid minimum value, labels, momenta, and count,
`count_periodic_symbol_minima`	finite-grid minima counter.

This list is a coverage boundary, not a generator-algebra interface: no helper currently constructs a finite coordinate generator, a finite boost, a finite rotation, or their commutator table.

29.8 Next Numerical Milestones

Proposed next suite additions, none claimed as achieved here, are:

1. systematic sampled residual scans on shrinking low-energy windows,
2. comparison of open-boundary first moments with periodic symbol tests.

Any finite periodic coordinate-generator version of the first-moment comparison requires a branch or Fourier-interpolation convention before it can be promoted beyond a local experiment.

30 Gaussian Boson Real-Space Energy Density

30.1 Status, Sources, and Dependencies

Status: convention, not a theorem. This shard fixes how the scalar Gaussian Hamiltonian is split into site-attached real-space energy densities. It preserves the total Hamiltonian used in the sourced nearest-neighbour scalar model and in CA-24–CA-28, but it does not claim a continuum tensor object, a conservation law, or covariance.

Dependencies: CA-23–CA-28 and CONVENTIONS.md (j), (l).

30.2 Cell Volume and Notation

Let the hypercubic lattice spacing be ϵ, so one spatial cell has volume ϵ^d. In this shard h_x is the physical energy density attached to the cell at x:

$$H_\epsilon = \epsilon^d \sum_x h_x.$$

The integrated site energy is

$$e_x := \epsilon^d h_x, \qquad H_\epsilon = \sum_x e_x.$$

This distinction matters because the generic first-moment notation of CONVENTIONS.md (h) writes $H = \sum_x h_x$; for the Gaussian block, the object to insert there is e_x, not the density h_x.

For fields Φ_x, Π_x and a real symmetric translation-invariant kernel $V_r = V_{-r}$, use

$$H_\epsilon = \epsilon^d \left(\frac{1}{2} \sum_x \Pi_x^2 + \frac{1}{2} \sum_{x,r} \Phi_x V_r \Phi_{x+r} \right).$$

For the nearest-neighbour Klein-Gordon source, the renormalized coefficients are

$$V_0 = m^2 + \frac{2d}{\epsilon^2}, \qquad V_{\pm e_a} = -\frac{1}{\epsilon^2},$$

which give the dispersion recorded in CA-24.

30.3 Equal-Endpoint Bond Split

The chosen local density is

$$h_x = \frac{1}{2}\Pi_x^2 + \frac{1}{2}V_0\Phi_x^2 + \frac{1}{4}\sum_{r\neq 0} V_r\big(\Phi_x\Phi_{x+r} + \Phi_{x-r}\Phi_x\big).$$

Each nonzero displacement bond is assigned half to each endpoint cell. Summing over x gives

$$\sum_x h_x = \frac{1}{2}\sum_x \Pi_x^2 + \frac{1}{2}\sum_{x,r} \Phi_x V_r \Phi_{x+r},$$

so this convention reproduces the total scalar Hamiltonian above after the cell-volume factor is restored.

The q, p notation in CA-24–CA-28 absorbs the cell volume into the coefficient bookkeeping. In that notation the same displayed formula defines e_x, and the physical density is $h_x = e_x/\epsilon^d$.

30.4 Boundary Policy

The default exact identities are infinite-lattice identities with summable tails, or finite periodic total-energy identities. On a finite open region, the density is obtained from the finite Hamiltonian actually named: include only bonds present in that Hamiltonian, and share each included bond equally between its included endpoints. No exterior ghost sites, omitted half-bonds, or boundary counterterms are implicit.

For periodic finite lattices, wrap-around bonds are included in the total energy split. This does not create a periodic position convention. Any periodic first moment still needs the branch, sawtooth, or interpolation policy required by CONVENTIONS.md (j).

30.5 Vacuum-Energy Policy

The density above is bare. No normal-ordering or vacuum subtraction is implicit. Once a state ω is named, the expectation-subtracted version may be recorded as

$$:h_x:_\omega := h_x - \omega(h_x)\mathbf{1}.$$

This changes H_ϵ by a scalar and therefore drops out of commutators, but it matters for absolute energies, expectation values, and Fourier modes of quadratic densities. The local subtraction source is the free-fermion Koo-Saleur setting, where the scaling-limit representation relies on subtracting vacuum or ground-state expectation values; this shard only records the analogous policy needed before such quantities are used in the scalar Gaussian block.

30.6 Improvement Policy

The default convention has no divergence improvement:

$$h'_x = h_x$$

unless a later shard explicitly names an edge field b, a discrete divergence $(\nabla_\epsilon \cdot b)_x$, and a boundary convention. Such a rewrite

$$h'_x = h_x + (\nabla_\epsilon \cdot b)_x$$

may leave the total energy unchanged on a periodic lattice or for decaying fields, but it is not harmless for position-weighted generators. A discrete summation-by-parts calculation shifts first moments by an edge-field term plus boundary contributions. For example, in one dimension with

$$(\nabla_\epsilon b)_n = \frac{b_{n+1/2} - b_{n-1/2}}{\epsilon}, \qquad x_n = \epsilon n,$$

one has

$$\epsilon \sum_n x_n (\nabla_\epsilon b)_n = -\epsilon \sum_n b_{n+1/2} + \text{boundary terms}.$$

Therefore improvements are separate data, not gauge choices to be silently modded out.

30.7 Compiler Consequence

A scalar Gaussian real-space generator proposal must now carry the tuple

$$\big(\epsilon^d,\ h_x,\ e_x = \epsilon^d h_x,\ \text{boundary policy},\ \text{subtraction policy},\ \text{improvement policy}\big).$$

The total symbol $\omega(k)^2$ remains sufficient for CA-24–CA-28, but it is not sufficient once a construction uses position weights or local quadratic modes. Those later constructions must cite this density convention or perform an explicit convention sweep.